

IMDB Sentiment Analysis with Word Embeddings

Binary classification of movie reviews (positive vs. negative) using the raw IMDB text corpus. The pipeline compares pre-trained GloVe embeddings against task-tuned embeddings, evaluates embedding quality with sentiment word lists, and builds a dual-branch (Y-network) LSTM + Conv1D model.

Goal: exceed 90% test accuracy through embedding fine-tuning, architecture search, and staged training.

Data Requirements

Place the following in the working directory:

- `aclImdb/` (IMDB review corpus)
- `glove.6B.300d.txt` (GloVe 300-d vectors)
- `positive.txt` / `negative.txt` (sentiment lexicons)

```
In [ ]: from tensorflow import keras  
keras.__version__
```

```
Out[ ]: '3.10.0'
```

2. Sentiment Lexicons

Load curated lists of positive and negative words used to evaluate embedding quality.

```
In [ ]: filepath = 'positive.txt'  
L_pos = []  
with open(filepath) as fp:  
    line = fp.readline()  
    cnt = 1  
    while line:  
        if cnt > 35:  
            L_pos.append(line.strip())  
            line = fp.readline()  
            cnt += 1  
for word in L_pos:  
    print(word)  
print(len(L_pos))
```

a+
abound
abounds
abundance
abundant
accessible
accessible
acclaim
acclaimed
acclamation
accolade
accolades
accommodative
accomodative
accomplish
accomplished
accomplishment
accomplishments
accurate
accurately
achievable
achievement
achievements
achievable
acumen
adaptable
adaptive
adequate
adjustable
admirable
admirably
admiration
admire
admirer
admiring
admiringly
adorable
adore
adored
adorer
adoring
adoringly
adroit
adroitly
adulate
adulation
adulatory
advanced
advantage
advantageous
advantageously
advantages
adventuresome
adventurous
advocate
advocated

advocates
affability
affable
affably
affectation
affection
affectionate
affinity
affirm
affirmation
affirmative
affluence
affluent
afford
affordable
affordably
afordable
agile
agilely
agility
agreeable
agreeableness
agreeably
all-around
alluring
alluringly
altruistic
altruistically
amaze
amazed
amazement
amazes
amazing
amazingly
ambitious
ambitiously
ameliorate
amenable
amenity
amiability
amiably
amiable
amicability
amicable
amicably
amity
ample
amply
amuse
amusing
amusingly
angel
angelic
apotheosis
appeal
appealing

applaud
appreciable
appreciate
appreciated
appreciates
appreciative
appreciatively
appropriate
approval
approve
ardent
ardently
ardor
articulate
aspiration
aspirations
aspire
assurance
assurances
assure
assuredly
assuring
astonish
astonished
astonishing
astonishingly
astonishment
astound
astounded
astounding
astoundingly
astutely
attentive
attraction
attractive
attractively
attune
audible
audibly
auspicious
authentic
authoritative
autonomous
available
aver
avid
avidly
award
awarded
awards
awe
awed
awesome
awesomely
awesomeness
awestruck

awsome
backbone
balanced
bargain
beauteous
beautiful
beautifullly
beautifully
beautify
beauty
beckon
beckoned
beckoning
beckons
believable
believeable
beloved
benefactor
beneficent
beneficial
beneficially
beneficiary
benefit
benefits
benevolence
benevolent
benifits
best
best-known
best-performing
best-selling
better
better-known
better-than-expected
beutifully
blameless
bless
blessing
bliss
blissful
blissfully
blithe
blockbuster
bloom
blossom
bolster
bonny
bonus
bonuses
boom
booming
boost
boundless
bountiful
brainiest
brainy

brand-new
brave
bravery
bravo
breakthrough
breakthroughs
breathlessness
breathtaking
breathtakingly
breeze
bright
brighten
brighter
brightest
brilliance
brilliances
brilliant
brilliantly
brisk
brotherly
bullish
buoyant
cajole
calm
calming
calmness
capability
capable
capably
captivate
captivating
carefree
cashback
cashbacks
catchy
celebrate
celebrated
celebration
celebratory
champ
champion
charisma
charismatic
charitable
charm
charming
charmingly
chaste
cheaper
cheapest
cheer
cheerful
cheery
cherish
cherished
cherub

chic
chivalrous
chivalry
civility
civilize
clarity
classic
classy
clean
cleaner
cleanest
cleanliness
cleanly
clear
clear-cut
cleared
clearer
clearly
clears
clever
cleverly
cohere
coherence
coherent
cohesive
colorful
comely
comfort
comfortable
comfortably
comforting
comfy
commend
commendable
commendably
commitment
commodious
compact
compactly
compassion
compassionate
compatible
competitive
complement
complementary
complemented
complements
compliant
compliment
complimentary
comprehensive
conciliate
conciliatory
concise
confidence
confident

congenial
congratulate
congratulation
congratulations
congratulatory
conscientious
considerate
consistent
consistently
constructive
consummate
contentment
continuity
contrasty
contribution
convenience
convenient
conveniently
convience
convienient
convient
convincing
convincingly
cool
coolest
cooperative
cooperatively
cornerstone
correct
correctly
cost-effective
cost-saving
counter-attack
counter-attacks
courage
courageous
courageously
courageousness
courteous
courtly
covenant
cozy
creative
credence
credible
crisp
crisper
cure
cure-all
cushy
cute
cuteness
danke
danken
daring
daringly

darling
dashing
dauntless
dawn
dazzle
dazzled
dazzling
dead-cheap
dead-on
decency
decent
decisive
decisiveness
dedicated
defeat
defeated
defeating
defeats
defender
deference
deft
deginified
delectable
delicacy
delicate
delicious
delight
delighted
delightful
delightfully
delightfulness
dependable
dependably
deservedly
deserving
desirable
desiring
desirous
destiny
detachable
devout
dexterous
dexterously
dextrous
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dignity
diligence
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diligently
diplomatic
dirt-cheap
distinction
distinctive
distinguished
diversified

divine
divinely
dominate
dominated
dominates
dote
dotingly
doubtless
dreamland
dumbfounded
dumbfounding
dummy-proof
durable
dynamic
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eagerly
eagerness
earnest
earnestly
earnestness
ease
eased
eases
easier
easiest
easiness
easing
easy
easy-to-use
easygoing
ebullience
ebullient
ebulliently
ecenomical
economical
ecstasies
ecstasy
ecstatic
ecstatically
edify
educated
effective
effectively
effectiveness
effectual
efficacious
efficient
efficiently
effortless
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elate

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enjoys
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enough

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enrichment
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entertaining
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enthusiast
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enticingly
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entrust
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enviably
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enviously
enviousness
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err-free
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evenly
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evocative
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fairly
fairness

faith
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faithfulness
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fanfare
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fascination
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fashionably
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fast-paced
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fastest
fastest-growing
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fave
favor
favorable
favored
favorite
favorited
favour
fearless
fearlessly
feasible
feasibly
feat
feature-rich
fecilitous
feisty
felicitate
felicitous
felicity
fertile
fervent
fervently
fervid
fervidly
fervor
festive
fidelity
fiery

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fine-looking
finely
finer
finest
firmer
first-class
first-in-class
first-rate
flashy
flatter
flattering
flatteringly
flawless
flawlessly
flexibility
flexible
flourish
flourishing
fluent
flutter
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fondly
fondness
foolproof
foremost
foresight
formidable
fortitude
fortuitous
fortuitously
fortunate
fortunately
fortune
fragrant
free
freed
freedom
freedoms
fresh
fresher
freshest
friendliness
friendly
frolic
frugal
fruitful
ftw
fulfillment
fun
futurestic
futuristic
gaiety
gaily
gain
gained

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gaining
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genius
gentle
gentlest
genuine
gifted
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gladden
gladly
gladness
glamorous
glee
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gleefully
glimmer
glimmering
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glistening
glitter
glitz
glorify
glorious
gloriously
glory
glow
glowing
glowingly
god-given
god-send
godlike
godsend
gold
golden
good
goodly
goodness
goodwill
goood
gooodd
gorgeous
gorgeously
grace

graceful
gracefully
gracious
graciously
graciousness
grand
grandeur
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gratefully
gratification
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gratifies
gratify
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gratifyingly
gratitude
great
greatest
greatness
grin
groundbreaking
guarantee
guidance
guiltless
gumption
gush
gusto
gutsy
hail
halcyon
hale
hallmark
hallmarks
hallowed
handier
handily
hands-down
handsome
handsomely
handy
happier
happily
happiness
happy
hard-working
hardier
hardy
harmless
harmonious
harmoniously
harmonize
harmony
headway
heal
healthful
healthy

hearten
heartening
heartfelt
heartily
heartwarming
heaven
heavenly
helped
helpful
helping
hero
heroic
heroically
heroine
heroize
heros
high-quality
high-spirited
hilarious
holy
homage
honest
honesty
honor
honorable
honored
honoring
hooray
hopeful
hospitable
hot
hotcake
hotcakes
hottest
hug
humane
humble
humility
humor
humorous
humorously
humour
humourous
ideal
idealize
ideally
idol
idolize
idolized
idyllic
illuminate
illuminati
illuminating
illumine
illustrious
ilu

imaculate
imaginative
immaculate
immaculately
immense
impartial
impartiality
impartially
impassioned
impeccable
impeccably
important
impress
impressed
impresses
impressive
impressively
impressiveness
improve
improved
improvement
improvements
improves
improving
incredible
incredibly
indebted
individualized
indulgence
indulgent
industrious
inestimable
inestimably
inexpensive
infallibility
infallible
infallibly
influential
ingenious
ingeniously
ingenuity
ingenuous
ingenuously
innocuous
innovation
innovative
inpressed
insightful
insightfully
inspiration
inspirational
inspire
inspiring
instantly
instructive
instrumental

integral
integrated
intelligence
intelligent
intelligible
interesting
interests
intimacy
intimate
intricate
intrigue
intriguing
intriguingly
intuitive
invaluable
invaluably
inventive
invigorate
invigorating
invincibility
invincible
inviolable
inviolable
invulnerable
irreplaceable
irreproachable
irresistible
irresistibly
issue-free
jaw-dropping
jaw-dropping
jollify
jolly
jovial
joy
joyful
joyfully
joyous
joyously
jubilant
jubilantly
jubilate
jubilation
jubilant
judicious
justly
keen
keenly
keenness
kid-friendly
kindliness
kindly
kindness
knowledgeable
kudos
large-capacity

laud
laudable
laudably
lavish
lavishly
law-abiding
lawful
lawfully
lead
leading
leads
lean
led
legendary
leverage
levity
liberate
liberation
liberty
lifesaver
light-hearted
lighter
likable
like
liked
likes
liking
lionhearted
lively
logical
long-lasting
lovable
lovably
love
loved
loveliness
lovely
lover
loves
loving
low-cost
low-price
low-priced
low-risk
lower-priced
loyal
loyalty
lucid
lucidly
luck
luckier
luckiest
luckiness
lucky
lucrative
luminous

lush
luster
lustrous
luxuriant
luxuriate
luxurious
luxuriously
luxury
lyrical
magic
magical
magnanimous
magnanimously
magnificence
magnificent
magnificently
majestic
majesty
manageable
maneuverable
marvel
marveled
marvelled
marvellous
marvelous
marvelously
marvelousness
marvels
master
masterful
masterfully
masterpiece
masterpieces
masters
mastery
matchless
mature
maturely
maturity
meaningful
memorable
merciful
mercifully
mercy
merit
meritorious
merrily
merriment
merriness
merry
mesmerize
mesmerized
mesmerizes
mesmerizing
mesmerizingly
meticulous

meticulously
mightily
mighty
mind-blowing
miracle
miracles
miraculous
miraculously
miraculousness
modern
modest
modesty
momentous
monumental
monumentally
morality
motivated
multi-purpose
navigable
neat
neatest
neatly
nice
nicely
nicer
nicest
nifty
nimble
noble
nobly
noiseless
non-violence
non-violent
notably
noteworthy
nourish
nourishing
nourishment
novelty
nurturing
oasis
obsession
obsessions
obtainable
openly
openness
optimal
optimism
optimistic
opulent
orderly
originality
outdo
outdone
outperform
outperformed

outperforming
outperforms
outshine
outshone
outsmart
outstanding
outstandingly
outstrip
outwit
ovation
overjoyed
overtake
overtaken
overtakes
overtaking
overtook
overture
pain-free
painless
painlessly
palatial
pamper
pampered
pamperedly
pamperedness
pampers
panoramic
paradise
paramount
pardon
passion
passionate
passionately
patience
patient
patiently
patriot
patriotic
peace
peaceable
peaceful
peacefully
peacekeepers
peach
peerless
pep
pepped
pepping
peppy
peps
perfect
perfection
perfectly
permissible
perseverance
persevere

personages
personalized
phenomenal
phenomenally
picturesque
piety
pinnacle
playful
playfully
pleasant
pleasantly
pleased
pleases
pleasing
pleasingly
pleasurable
pleasurably
pleasure
plentiful
pluses
plush
plusses
poetic
poeticize
poignant
poise
poised
polished
polite
politeness
popular
portable
posh
positive
positively
positives
powerful
powerfully
praise
praiseworthy
praising
pre-eminent
precious
precise
precisely
preeminent
prefer
preferable
preferably
preferred
prefers
preferring
prefers
premier
prestige
prestigious

prettily
pretty
priceless
pride
principled
privilege
privileged
prize
proactive
problem-free
problem-solver
prodigious
prodigiously
prodigy
productive
productively
proficient
proficiently
profound
profoundly
profuse
profusion
progress
progressive
prolific
prominence
prominent
promise
promised
promises
promising
promoter
prompt
promptly
proper
properly
propitious
propitiously
pros
prosper
prosperity
prosperous
prospros
protect
protection
protective
proud
proven
proves
providence
proving
prowess
prudence
prudent
prudently
punctual

pure
purify
purposeful
quaint
qualified
qualify
quicker
quiet
quieter
radiance
radiant
rapid
rapport
rapt
rapture
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raptureously
rapturous
rapturously
rational
razor-sharp
reachable
readable
readily
ready
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reaffirmation
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realizable
reasonable
reasonably
reasoned
reassurance
reassure
receptive
reclaim
recomend
recommend
recommendation
recommendations
recommended
reconcile
reconciliation
record-setting
recover
recovery
rectification
rectify
rectifying
redeem
redeeming
redemption
refine
refined
refinement
reform

reformed
reforming
reforms
refresh
refreshed
refreshing
refund
refunded
regal
regally
regard
rejoice
rejoicing
rejoicingly
rejuvenate
rejuvenated
rejuvenating
relaxed
relent
reliable
reliably
relief
relish
remarkable
remarkably
remedy
remission
remunerate
renaissance
renewed
renown
renowned
replaceable
reputable
reputation
resilient
resolute
resound
resounding
resourceful
resourcefulness
respect
respectable
respectful
respectfully
respite
resplendent
responsibly
responsive
restful
restored
restructure
restructured
restructuring
retractable
revel

revelation
revere
reverence
reverent
reverently
revitalize
revival
revive
revives
revolutionary
revolutionize
revolutionized
revolutionizes
reward
rewarding
rewardingly
rich
richer
richly
richness
right
righten
righteous
righteously
righteousness
rightful
rightfully
rightly
rightness
risk-free
robust
rock-star
rock-stars
rockstar
rockstars
romantic
romantically
romanticize
roomier
roomy
rosy
safe
safely
sagacity
sagely
saint
saintliness
saintly
salutary
salute
sane
satisfactorily
satisfactory
satisfied
satisfies
satisfy

satisfying
satisfied
saver
savings
savior
savvy
scenic
seamless
seasoned
secure
securely
selective
self-determination
self-respect
self-satisfaction
self-sufficiency
self-sufficient
sensation
sensational
sensationally
sensations
sensible
sensibly
sensitive
serene
serenity
sexy
sharp
sharper
sharpest
shimmering
shimmeringly
shine
shiny
significant
silent
simpler
simplest
simplified
simplifies
simplify
simplifying
sincere
sincerely
sincerity
skill
skilled
skillful
skillfully
slammin
sleek
slick
smart
smarter
smartest
smartly

smile
smiles
smiling
smilingly
smitten
smooth
smoother
smoothes
smoothest
smoothly
snappy
snazzy
sociable
soft
softer
solace
solicitous
solicitously
solid
solidarity
soothe
soothingly
sophisticated
soulful
soundly
soundness
spacious
sparkle
sparkling
spectacular
spectacularly
speedily
speedy
spellbind
spellbinding
spellbindingly
spellbound
spirited
spiritual
splendid
splendidly
splendor
spontaneous
sporty
spotless
sprightly
stability
stabilize
stable
stainless
standout
state-of-the-art
stately
statuesque
staunch
staunchly

staunchness
steadfast
steadfastly
steadfastness
steadiest
steadiness
steady
stellar
stellarly
stimulate
stimulates
stimulating
stimulative
stirringly
straighten
straightforward
streamlined
striking
strikingly
striving
strong
stronger
strongest
stunned
stunning
stunningly
stupendous
stupendously
sturdier
sturdy
stylish
stylishly
stylized
suave
suavely
sublime
subsidize
subsidized
subsidizes
subsidizing
substantive
succeed
succeeded
succeeding
succeeds
succes
success
successes
successful
successfully
suffice
sufficed
suffices
sufficient
sufficiently
suitable

sumptuous
sumptuously
sumptuousness
super
superb
superbly
superior
superiority
supple
support
supported
supporter
supporting
supportive
supports
supremacy
supreme
supremely
supurb
supurbly
surmount
surpass
surreal
survival
survivor
sustainability
sustainable
swank
swankier
swankiest
swanky
sweeping
sweet
sweeten
sweetheart
sweetly
sweetness
swift
swiftness
talent
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talents
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tenaciously
tenacity
tender
tenderly
terrific
terrifically
thank

thankful
thinner
thoughtful
thoughtfully
thoughtfulness
thrift
thrifty
thrill
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thrilling
thrillingly
thrills
thrive
thriving
thumb-up
thumbs-up
tickle
tidy
time-honored
timely
tingle
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titillatingly
togetherness
tolerable
toll-free
top
top-notch
top-quality
topnotch
tops
tough
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toughest
traction
tranquil
tranquility
transparent
treasure
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trendy
triumph
triumphal
triumphant
triumphantly
trivially
trophy
trouble-free
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trumpet
trust
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trustingly
trustworthiness

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trusty
truthful
truthfully
truthfulness
twinkly
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unabashedly
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unassailable
unbeatable
unbiased
unbound
uncomplicated
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undaunted
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unencumbered
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unquestionable
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unrestricted
unrivalled
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upliftment
upscale
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user-friendly
user-replaceable
valiant

valiantly
valor
valuable
variety
venerate
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veritable
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versatility
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vibrantly
victorious
victory
viewable
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vigilant
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virtuously
visionary
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vouchsafe
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warmly
warmth
wealthy
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well-positioned
well-received
well-regarded
well-rounded
well-run
well-wishers
wellbeing
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wholeheartedly
wholesome

whoaa
whoaaa
wieldy
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willingly
willingness
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winner
winners
winning
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wisdom
wise
wisely
witty
won
wonder
wonderful
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wondrous
woo
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workable
worked
works
world-famous
worth
worth-while
worthiness
worthwhile
worthy
wow
wowed
wowing
wows
yay
youthful
zeal
zenith
zest
zippy
2006

```
In [ ]: filepath = 'negative.txt'
        L_neg = []
        with open(filepath) as fp:
            line = fp.readline()
            cnt = 1
            while line:
                if cnt > 35:
                    L_neg.append(line.strip())
```

```
        line = fp.readline()
        cnt += 1
    for word in L_neg:
        print(word)
    print(len(L_neg))
```

2-faced
2-faces
abnormal
abolish
abominable
abominably
abominate
abomination
abort
aborted
aborts
abrade
abrasive
abrupt
abruptly
abscond
absence
absent-minded
absentee
absurd
absurdity
absurdly
absurdness
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abuses
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abysmal
abysmally
abyss
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accost
accursed
accusation
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acerbate
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acerbically
ache
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acrimony
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adversity
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affliction
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affront
afraid
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aggravation
aggression
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aggressiveness
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aloof

altercation
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ambivalent
ambush
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anguish
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anti-proliferation
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averse

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dictator

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exhort
exile
exorbitant
exorbitantance
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expire
expired
explode
exploit
exploitation
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fails

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failures
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falsehood
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plight
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plotters
ploy
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plunderer
pointless
pointlessly
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poisonous
poisonously
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poky
polarisation
polemize
pollute
polluter
polluters
polution
pompous
poor
poorer
poorest
poorly
posturing
pout
poverty
powerless
prate
pratfall
prattle
precarious
precariously
precipitate
precipitous
predatory
predicament
prejudge
prejudice
prejudices
prejudicial
premeditated
preoccupy
preposterous
preposterously

presumptuous
presumptuously
pretence
pretend
pretense
pretentious
pretentiously
prevaricate
pricey
pricier
prick
prickle
prickles
prideful
prik
primitive
prison
prisoner
problem
problematic
problems
procrastinate
procrastinates
procrastination
profane
profanity
prohibit
prohibitive
prohibitively
propaganda
propagandize
proprietary
prosecute
protest
protested
protesting
protests
protracted
provocation
provocative
provoke
pry
pugnacious
pugnaciously
pugnacity
punch
punish
punishable
punitive
punk
puny
puppet
puppets
puzzled
puzzlement
puzzling

quack
qualm
qualms
quandary
quarrel
quarrellous
quarrellously
quarrels
quarrelsome
quash
queer
questionable
quibble
quibbles
quitter
rabid
racism
racist
racists
racy
radical
radicalization
radically
radicals
rage
ragged
raging
rail
raked
rampage
rampant
ramshackle
rancor
randomly
rankle
rant
ranted
ranting
rantingly
rants
rape
raped
raping
rascal
rascals
rash
rattle
rattled
rattles
ravage
raving
reactionary
rebellious
rebuff
rebuke
recalcitrant

recant
recession
recessionary
reckless
recklessly
recklessness
recoil
recourses
redundancy
redundant
refusal
refuse
refused
refuses
refusing
refutation
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refuted
refutes
refuting
regress
regression
regressive
regret
regreted
regretful
regretfully
regrets
regrettable
regrettably
regretted
reject
rejected
rejecting
rejection
rejects
relapse
relentless
relentlessly
relentlessness
reluctance
reluctant
reluctantly
remorse
remorseful
remorsefully
remorseless
remorselessly
remorselessness
renounce
renunciation
repel
repetitive
reprehensible
reprehensibly
reprehension

reprehensive
repress
repression
repressive
reprimand
reproach
reproachful
reprove
reprovingly
repudiate
repudiation
repugn
repugnance
repugnant
repugnantly
repulse
repulsed
repulsing
repulsive
repulsively
repulsiveness
resent
resentful
resentment
resignation
resigned
resistance
restless
restlessness
restrict
restricted
restriction
restrictive
resurgent
retaliate
retaliatory
retard
retarded
retardedness
retards
reticent
retract
retreat
retreated
revenge
revengeful
revengefully
revert
revile
reviled
revoke
revolt
revolting
revoltingly
revulsion
revulsive

rhapsodize
rhetoric
rhetorical
ricer
ridicule
ridicules
ridiculous
ridiculously
rife
rift
rifts
rigid
rigidity
rigidness
rile
riled
rip
rip-off
ripcoff
ripped
risk
risks
risky
rival
rivalry
roadblocks
rocky
rogue
rollercoaster
rot
rotten
rough
rremediable
rubbish
rude
rue
ruffian
ruffle
ruin
ruined
ruining
ruinous
ruins
rumbling
rumor
rumors
rumours
rumple
run-down
runaway
rupture
rust
rusts
rusty
rut
ruthless

ruthlessly
ruthlessness
ruts
sabotage
sack
sacrificed
sad
sadden
sadly
sadness
sag
sagged
sagging
saggy
sags
salacious
sanctimonious
sap
sarcasm
sarcastic
sarcastically
sardonic
sardonically
sass
satirical
satirize
savage
savaged
savagery
savages
scaly
scam
scams
scandal
scandalize
scandalized
scandalous
scandalously
scandals
scandel
scandels
scant
scapegoat
scar
scarce
scarcely
scarcity
scare
scared
scarier
scariest
scarily
scarred
scars
scary
scathing

scathingly
sceptical
scoff
scoffingly
scold
scolded
scolding
scoldingly
scorching
scorchingly
scorn
scornful
scornfully
scoundrel
scourge
scowl
scramble
scrambled
scrambles
scrambling
scrap
scratch
scratched
scratches
scratchy
scream
screech
screw-up
screwed
screwed-up
screwy
scuff
scuffs
scum
scummy
second-class
second-tier
secretive
sedentary
seedy
seethe
seething
self-coup
self-criticism
self-defeating
self-destructive
self-humiliation
self-interest
self-interested
self-serving
selfinterested
selfish
selfishly
selfishness
semi-retarded
senile

sensationalize
senseless
senselessly
seriousness
sermonize
servitude
set-up
setback
setbacks
sever
severe
severity
sh*t
shabby
shadowy
shady
shake
shaky
shallow
sham
shambles
shame
shameful
shamefully
shamefulness
shameless
shamelessly
shamelessness
shark
sharply
shatter
shemale
shimmer
shimmy
shipwreck
shirk
shirker
shit
shiver
shock
shocked
shocking
shockingly
shoddy
short-lived
shortage
shortchange
shortcoming
shortcomings
shortness
shortsighted
shortsightedness
showdown
shrew
shriek
shrill

shrilly
shrivel
shroud
shrouded
shrug
shun
shunned
sick
sicken
sickening
sickeningly
sickly
sickness
sidetrack
sidetracked
siege
sillily
silly
simplistic
simplistically
sin
sinful
sinfully
sinister
sinisterly
sink
sinking
skeletons
skeptic
skeptical
skeptically
skepticism
sketchy
skimpy
skinny
skittish
skittishly
skulk
slack
slander
slanderer
slanderous
slanderously
slanders
slap
slashing
slaughter
slaughtered
slave
slaves
sleazy
slime
slog
slogged
slogging
slogs

sloooooooooooooooooow
sloooooow
sloooow
sloow
sloppily
sloppy
sloth
slothful
slow
slow-moving
slowed
slower
slowest
slowly
sloww
slowww
slowwww
slug
sluggish
slump
slumping
slumpping
slur
slut
sluts
sly
smack
smallish
smash
smear
smell
smelled
smelling
smells
smelly
smelt
smoke
smokescreen
smolder
smoldering
smother
smoulder
smouldering
smudge
smudged
smudges
smudging
smug
smugly
smut
smuttier
smuttiest
smutty
snag
snagged
snagging

snags
snappish
snappishly
snare
snarky
snarl
sneak
sneakily
sneaky
sneer
sneering
sneeringly
snob
snobbish
snobby
snobish
snobs
snub
so-cal
soapy
sob
sober
sobering
solemn
solicitude
somber
sore
sorely
soreness
sorrow
sorrowful
sorrowfully
sorry
sour
sourly
spade
spank
spendy
spew
spewed
spewing
spews
spilling
spinster
spiritless
spite
spiteful
spitefully
spitefulness
splatter
split
splitting
spoil
spoilage
spoilages
spoiled

spoiled
spoils
spook
spookier
spookiest
spookily
spooky
spoon-fed
spoon-feed
spoonfed
sporadic
spotty
spurious
spurn
sputter
squabble
squabbling
squander
squash
squeak
squeaks
squeaky
squeal
squealing
squeals
squirm
stab
stagnant
stagnate
stagnation
staid
stain
stains
stale
stalemate
stall
stalls
stammer
stampede
standstill
stark
starkly
startle
startling
startlingly
starvation
starve
static
steal
stealing
steals
steep
steeply
stench
stereotype
stereotypical

stereotypically
stern
stew
sticky
stiff
stiffness
stifle
stifling
stiflingly
stigma
stigmatize
sting
stinging
stingingly
stingy
stink
stinks
stodgy
stole
stolen
stooge
stooges
stormy
straggle
straggler
strain
strained
straining
strange
strangely
stranger
strangest
strangle
streaky
strenuous
stress
stresses
stressful
stressfully
stricken
strict
strictly
strident
stridently
strife
strike
stringent
stringently
struck
struggle
struggled
struggles
struggling
strut
stubborn
stubbornly

stubbornness
stuck
stuffy
stumble
stumbled
stumbles
stump
stumped
stumps
stun
stunt
stunted
stupid
stupidest
stupidity
stupidly
stupified
stupify
stupor
stutter
stuttered
stuttering
stutters
sty
stymied
sub-par
subdued
subjected
subjection
subjugate
subjugation
submissive
subordinate
subpoena
subpoenas
subservience
subservient
substandard
subtract
subversion
subversive
subversively
subvert
succumb
suck
sucked
sucker
sucks
sucky
sue
sued
sueing
sues
suffer
suffered
sufferer

sufferers
suffering
suffers
suffocate
sugar-coat
sugar-coated
sugarcoated
suicidal
suicide
sulk
sullen
sully
sunder
sunk
sunken
superficial
superficiality
superficially
superfluous
superstition
superstitious
suppress
suppression
surrender
susceptible
suspect
suspicion
suspicious
suspiciously
swagger
swamped
sweaty
swelled
swelling
swindle
swipe
swollen
symptom
symptoms
syndrome
taboo
tacky
taint
tainted
tamper
tangle
tangled
tangles
tank
tanked
tanks
tantrum
tardy
tarnish
tarnished

tarnishes
tarnishing
tattered
taunt
taunting
tauntingly
taunts
taut
tawdry
taxing
tease
teasingly
tedious
tediously
temerity
temper
tempest
temptation
tenderness
tense
tension
tentative
tentatively
tenuous
tenuously
tepid
terrible
terribleness
terribly
terror
terror-genic
terrorism
terrorize
testily
testy
tetchily
tetchy
thankless
thicker
thirst
thorny
thoughtless
thoughtlessly
thoughtlessness
thrash
threat
threaten
threatening
threats
threesome
throb
throbbled
throbbing
throbs
throttle
thug

thumb-down
thumbs-down
thwart
time-consuming
timid
timidity
timidly
timidness
tin-y
tingled
tingling
tired
tiresome
tiring
tiringly
toil
toll
top-heavy
topple
torment
tormented
torrent
tortuous
torture
tortured
tortures
torturing
torturous
torturously
totalitarian
touchy
toughness
tout
touted
touts
toxic
traduce
tragedy
tragic
tragically
traitor
traitorous
traitorously
tramp
trample
transgress
transgression
trap
traped
trapped
trash
trashed
trashy
trauma
traumatic
traumatically

traumatize
traumatized
travesties
travesty
treacherous
treacherously
treachery
treason
treasonous
trick
tricked
trickery
tricky
trivial
trivialize
trouble
troubled
troublemaker
troubles
troublesome
troublesomely
troubling
troublingly
truant
tumble
tumbled
tumbles
tumultuous
turbulent
turmoil
twist
twisted
twists
two-faced
two-faces
tyrannical
tyrannically
tyranny
tyrant
ugh
uglier
ugliest
ugliness
ugly
ulterior
ultimatum
ultimatums
ultra-hardline
un-viewable
unable
unacceptable
unacceptably
unacceptably
unacceptable
unaccustomed
unachievable

unaffordable
unappealing
unattractive
unauthentic
unavailable
unavoidably
unbearable
unbearably
unbelievable
unbelievably
uncaring
uncertain
uncivil
uncivilized
unclean
unclear
uncollectible
uncomfortable
uncomfortably
uncomfy
uncompetitive
uncompromising
uncompromisingly
unconfirmed
unconstitutional
uncontrolled
unconvincing
unconvincingly
uncooperative
uncouth
uncreative
undecided
undefined
undependability
undependable
undercut
undercuts
undercutting
underdog
underestimate
underlings
undermine
undermined
undermines
undermining
underpaid
underpowered
undersized
undesirable
undetermined
undid
undignified
undissolved
undocumented
undone
undue

unease
uneasily
uneasiness
uneasy
uneconomical
unemployed
unequal
unethical
uneven
uneventful
unexpected
unexpectedly
unexplained
unfairly
unfaithful
unfaithfully
unfamiliar
unfavorable
unfeeling
unfinished
unfit
unforeseen
unforgiving
unfortunate
unfortunately
unfounded
unfriendly
unfulfilled
unfunded
ungovernable
ungrateful
unhappily
unhappiness
unhappy
unhealthy
unhelpful
unilateralism
unimaginable
unimaginably
unimportant
uninformed
uninsured
unintelligible
unintelligible
unipolar
unjust
unjustifiable
unjustifiably
unjustified
unjustly
unkind
unkindly
unknown
unlamentable
unlamentably
unlawful

unlawfully
unlawfulness
unleash
unlicensed
unlikely
unlucky
unmoved
unnatural
unnaturally
unnecessary
unneeded
unnerve
unnerved
unnerving
unnervingly
unnoticed
unobserved
unorthodox
unorthodoxy
unpleasant
unpleasantries
unpopular
unpredictable
unprepared
unproductive
unprofitable
unprove
unproved
unproven
unproves
unproving
unqualified
unravel
unraveled
unreachable
unreadable
unrealistic
unreasonable
unreasonably
unrelenting
unrelentingly
unreliability
unreliable
unresolved
unresponsive
unrest
unruly
unsafe
unsatisfactory
unsavory
unscrupulous
unscrupulously
unsecure
unseemly
unsettle
unsettled

unsettling
unsettlingly
unskilled
unsophisticated
unsound
unspeakable
unspeakably
unspecified
unstable
unsteadily
unsteadiness
unsteady
unsuccessful
unsuccessfully
unsupported
unsupportive
unsure
unsuspecting
unsustainable
untenable
untested
unthinkable
unthinkably
untimely
untouched
untrue
untrustworthy
untruthful
unusable
unusably
unuseable
unuseably
unusual
unusually
unviewable
unwanted
unwarranted
unwatchable
unwelcoming
unwell
unwieldy
unwilling
unwillingly
unwillingness
unwise
unwisely
unworkable
unworthy
unyielding
upbraid
upheaval
uprising
uproar
uproarious
uproariously
uproarious

uproarously
uproot
upset
upsetting
upsets
upsetting
upsettingly
urgent
useless
usurp
usurper
utterly
vagrant
vague
vagueness
vain
vainly
vanity
vehement
vehemently
vengeance
vengeful
vengefully
vengefulness
venom
venomous
venomously
vent
vestiges
vex
vexation
vexing
vexingly
vibrate
vibrated
vibrates
vibrating
vibration
vice
vicious
viciously
viciousness
victimize
vile
vileness
vilify
villainous
villainously
villains
villian
villianous
villianously
villify
vindictive
vindictively
vindictiveness

violate
violation
violator
violators
violent
violently
viper
virulence
virulent
virulently
virus
vociferous
vociferously
volatile
volatility
vomit
vomited
vomiting
vomits
vulgar
vulnerable
wack
wail
wallow
wane
waning
wanton
war-like
warily
wariness
warlike
warned
warning
warp
warped
wary
washed-out
waste
wasted
wasteful
wastefulness
wasting
water-down
watered-down
wayward
weak
weaken
weakening
weaker
weakness
weaknesses
weariness
wearisome
weary
wedge
weed

weep
weird
weirdly
wheedle
whimper
whine
whining
whiny
whips
whore
whores
wicked
wickedly
wickedness
wild
wildly
wiles
wilt
wily
wimpy
wince
wobble
wobbled
wobbles
woe
woebegone
woeful
woefully
womanizer
womanizing
worn
worried
worriedly
worrier
worries
worrisome
worry
worrying
worryingly
worse
worsen
worsening
worst
worthless
worthlessly
worthlessness
wound
wounds
wrangle
wrath
wreak
wreaked
wreaks
wreck
wrest
wrestle

wretch
wretched
wretchedly
wretchedness
wrinkle
wrinkled
wrinkles
wrip
wripped
wripping
writhe
wrong
wrongful
wrongly
wrought
yawn
zap
zapped
zaps
zealot
zealous
zealously
zombie
4783

3. Word Embeddings Overview

Two strategies: learn embeddings jointly with the classifier, or initialize from pre-trained vectors (GloVe) and optionally fine-tune.

3.1 Learning Embeddings In-Task

Random initialization works when enough data is available — the network learns structure during training.

3.2 Pre-Trained Embeddings (GloVe)

When data is limited, import 300-d GloVe vectors trained on 6B tokens to bootstrap semantic structure.

3.3 End-to-End Pipeline

Download raw IMDB reviews, tokenize, map to GloVe vectors, and train a classifier on top.

4. Load IMDB Reviews

Read all training reviews as raw text strings with binary labels.

```
In [ ]: import os
imdb_dir = 'aclImdb'
train_dir = os.path.join(imdb_dir, 'train')
labels = []
texts = []
for label_type in ['neg', 'pos']:
    dir_name = os.path.join(train_dir, label_type)
    for fname in os.listdir(dir_name):
        if fname[-4:] == '.txt':
            f = open(os.path.join(dir_name, fname))
            texts.append(f.read())
            f.close()
            if label_type == 'neg':
                labels.append(0)
            else:
                labels.append(1)
```

```
In [ ]: texts[:2]
```

```
Out[ ]: ['Warner Bros. made many potboilers in the 1930s and most of them are fast
paced, economical and very entertaining. I really love how the studio explo
ited the less glamorous elements of our daily life. This is one of Warner
\'s few hard-edged melodramas that simply doesn\'t work. Edward G. Robinson
plays a ruthless editor of newspaper who resurrects the 20-year old story o
f a murderess with tragic consequences. Robinson gives a lively performance
but he is surrounded by actors that don\'t cut the mustard. H.B. Warner, Al
ine MacMahon and Boris Karloff are good, but the bad acting of Frances Star
r and Anthony Bushell in the second leads really hurts the movie. Starr is
particularly bad during her big dramatic scene near the end of the movie. I
n addition, the moralistic tone of the film seems ridiculous in the context
of pre-code Hollywood. LeRoy\'s direction is full of innovative visual touc
hes but he cannot overcome the bad acting and the unintentionally funny sit
uations. Sol Polito\'s camera work is strong. Somehow, this piece of dreck
got an Oscar nomination for Best Picture (in a year that gave us "M," "Drac
ula" and "Frankenstein").',
```

```
'This film was bad because there was nothing interesting about it. It was
sort of a remake but then again, not really. I was very disappointed consid
ering the tools that Tim Burton has at his disposal. He had great make up a
nd CGI available and lots of money ($100 million) but can you honestly say
that what we got was as good as the original film that was made for less th
an $6 million? Heck no. So in that regard, the film fails.<br /><br />At le
ast in the original film, the statue of liberty scene was shocking but ther
e was nothing shocking in this film even the end scene because you could ki
nd of see it coming. And, by the way, if you give it some thought, and I di
d, you can figure out how the ending could come about. I just wish I had ba
ck the time that I wasted thinking about it.<br /><br />This film would hav
e been brilliant and fun if it stayed along the lines of the first film and
adopted part of "Beneath the Planet of the Apes". Here\'s how I would have
written it:<br /><br />An astronaut (it really should have been more than M
arky Mark because he\'s not good enough to care a film all by himself so I
would have put in three guys) that would land on what they would think woul
d be another planet, encounter humans (not mutants), a city ruled by apes, we
re hunted, made friends with some good apes, discover that they\'re on eart
h via finding some destroyed ruins, end up running into crazy mutant humans
living beneath the earth, a war breaks out between the mutant humans and th
e apes, and then....well, let your imagination run wild on how you want to
end it.<br /><br />But my point is that there could have been so much more
to this film. Sadly, Tim Burton really knows how to wreck a good thing. Con
sider what he did with the Batman series. He\'s a rotten director inspite w
hat of people think. He\'s trendy so he must be good goes the thinking. Sor
ry, but no. If anyone else had done this film, it would have turned out far
better and would have been far more satisfying.']
```

```
In [ ]: labels[:2]
```

```
Out[ ]: [0, 0]
```

```
In [ ]: len(labels)
```

```
Out[ ]: 25000
```

5. Tokenization

Convert text to padded integer sequences. Shuffle before splitting into 80/20 train/validation.

```
In [ ]: from tensorflow.keras.preprocessing.text import Tokenizer
        from tensorflow.keras.preprocessing.sequence import pad_sequences
        import numpy as np
        maxlen = 500
        training_samples = 20000
        validation_samples = 5000
        max_words = 10000
        tokenizer = Tokenizer(num_words=max_words)
        tokenizer.fit_on_texts(texts)
        sequences = tokenizer.texts_to_sequences(texts)
        word_index = tokenizer.word_index
        print('Found %s unique tokens.' % len(word_index))
        data = pad_sequences(sequences, maxlen=maxlen)
        labels = np.asarray(labels)
        print('Shape of data tensor:', data.shape)
        print('Shape of label tensor:', labels.shape)
        indices = np.arange(data.shape[0])
        np.random.shuffle(indices)
        data = data[indices]
        labels = labels[indices]
        x_train = data[:training_samples]
        y_train = labels[:training_samples]
        x_val = data[training_samples: training_samples + validation_samples]
        y_val = labels[training_samples: training_samples + validation_samples]
```

```
Found 88582 unique tokens.
Shape of data tensor: (25000, 500)
Shape of label tensor: (25000,)
```

```
In [ ]: word_index
```

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```

```
In [ ]: data.shape
```

```
Out[ ]: (25000, 500)
```

6. Load GloVe Embeddings

Parse `glove.6B.300d.txt` into a word → vector dictionary.

```
In [ ]: glove_dir = '.'
embeddings_index = {}
f = open(os.path.join(glove_dir, 'glove.6B.300d.txt'))
for line in f:
    values = line.split()
    word = values[0]
    coefs = np.asarray(values[1:], dtype='float32')
    embeddings_index[word] = coefs
f.close()
print('Found %s word vectors.' % len(embeddings_index))
```

Found 400000 word vectors.

Build the embedding matrix (`max_words × 300`) mapping tokenizer indices to GloVe vectors. Index 0 is reserved for padding/unknown.

```
In [ ]: embedding_dim = 300
embedding_matrix = np.zeros((max_words, embedding_dim))
for word, i in word_index.items():
    embedding_vector = embeddings_index.get(word)
    if i < max_words:
        if embedding_vector is not None:
            embedding_matrix[i] = embedding_vector
```

Filter positive/negative lexicon words to those present in both GloVe and the tokenizer vocabulary.

```
In [ ]: L_POS = []
for word in L_pos:
    if word in embeddings_index and word in word_index and word_index[word]:
        L_POS.append(word)
L_NEG = []
for word in L_neg:
    if word in embeddings_index and word in word_index and word_index[word]:
        L_NEG.append(word)
len(L_POS), len(L_NEG)
```

Out[]: (706, 1164)

Most frequent positive sentiment words (by token index) in the vocabulary.

```
In [ ]: top_pos = sorted(L_POS, key=lambda w: word_index[w])[:100]
print(top_pos)
```

```
['like', 'good', 'well', 'great', 'best', 'love', 'better', 'work', 'pretty', 'enough', 'right', 'interesting', 'fun', 'worth', 'beautiful', 'excellent', 'nice', 'top', 'classic', 'enjoy', 'recommend', 'wonderful', 'perfect', 'liked', 'entertaining', 'loved', 'fans', 'fine', 'amazing', 'humor', 'lead', 'works', 'enjoyed', 'favorite', 'brilliant', 'decent', 'strong', 'wonder', 'hero', 'hilarious', 'cool', 'happy', 'important', 'talent', 'modern', 'clearly', 'supporting', 'fast', 'romantic', 'enjoyable', 'easy', 'fantastic', 'clear', 'famous', 'realistic', 'leads', 'greatest', 'believable', 'free', 'hot', 'superb', 'memorable', 'beauty', 'perfectly', 'worked', 'incredibly', 'leading', 'powerful', 'masterpiece', 'fairly', 'talented', 'success', 'rich', 'cute', 'sweet', 'incredible', 'pure', 'popular', 'convincing', 'intelligent', 'clever', 'successful', 'positive', 'exciting', 'effective', 'amusing', 'appreciate', 'solid', 'impressive', 'respect', 'super', 'win', 'recommended', 'awesome', 'won', 'honest', 'tough', 'charming', 'likes', 'magic']
```

Most frequent negative sentiment words (by token index) in the vocabulary.

```
In [ ]: top_neg = sorted(L_NEG, key=lambda w: word_index[w])[:100]
        print(top_neg)
```

```
['bad', 'plot', 'funny', 'worst', 'hard', 'poor', 'death', 'dead', 'wrong', 'boring', 'awful', 'stupid', 'terrible', 'lost', 'worse', 'waste', 'problem', 'evil', 'killer', 'dark', 'unfortunately', 'kill', 'horrible', 'slow', 'killed', 'lack', 'murder', 'crap', 'hell', 'annoying', 'sad', 'scary', 'ridiculous', 'strange', 'disappointed', 'cheap', 'silly', 'problems', 'miss', 'falls', 'mystery', 'dull', 'hate', 'die', 'sorry', 'fall', 'weak', 'crime', 'doubt', 'lame', 'poorly', 'zombie', 'difficult', 'killing', 'shame', 'badly', 'weird', 'crazy', 'creepy', 'mess', 'cheesy', 'monster', 'joke', 'break', 'dumb', 'fails', 'twist', 'odd', 'sadly', 'cold', 'missed', 'wasted', 'revenge', 'tension', 'fear', 'kills', 'bored', 'trouble', 'violent', 'died', 'pointless', 'trash', 'bizarre', 'impossible', 'mad', 'prison', 'sick', 'disturbing', 'failed', 'fake', 'fiction', 'pathetic', 'garbage', 'utterly', 'attack', 'loud', 'unbelievable', 'twists', 'complex', 'laughable']
```

7. Baseline Model (Frozen GloVe)

Embedding → Flatten → Dense classifier. GloVe weights are frozen.

```
In [ ]: from keras.models import Sequential
        from keras.layers import Embedding, Flatten, Dense
        assert embedding_matrix.shape == (max_words, embedding_dim), "embedding_matrix shape mismatch"
        model = Sequential()
        model.add(Embedding(input_dim=max_words,
                             output_dim=embedding_dim,
                             weights=[embedding_matrix],
                             trainable=False))
        model.add(Flatten())
        model.add(Dense(32, activation='relu'))
        model.add(Dense(1, activation='sigmoid'))
```

Load pre-computed GloVe weights into the Embedding layer.

Train & Evaluate Baseline

```
In [ ]: model.compile(optimizer='rmsprop',
                      loss='binary_crossentropy',
                      metrics=['acc'])
history = model.fit(x_train, y_train,
                    epochs=10,
                    batch_size=32,
                    validation_data=(x_val, y_val))
```

Epoch 1/10

625/625  **8s** 6ms/step - acc: 0.5918 - loss: 0.7492 - val_acc: 0.7562 - val_loss: 0.5113

Epoch 2/10

625/625  **3s** 5ms/step - acc: 0.8565 - loss: 0.3472 - val_acc: 0.7694 - val_loss: 0.5234

Epoch 3/10

625/625  **5s** 4ms/step - acc: 0.9340 - loss: 0.1847 - val_acc: 0.7678 - val_loss: 0.6452

Epoch 4/10

625/625  **2s** 3ms/step - acc: 0.9709 - loss: 0.0914 - val_acc: 0.7592 - val_loss: 0.8270

Epoch 5/10

625/625  **2s** 3ms/step - acc: 0.9852 - loss: 0.0475 - val_acc: 0.7422 - val_loss: 1.1596

Epoch 6/10

625/625  **2s** 3ms/step - acc: 0.9913 - loss: 0.0278 - val_acc: 0.7508 - val_loss: 1.2166

Epoch 7/10

625/625  **2s** 3ms/step - acc: 0.9928 - loss: 0.0195 - val_acc: 0.7412 - val_loss: 1.4740

Epoch 8/10

625/625  **2s** 4ms/step - acc: 0.9954 - loss: 0.0138 - val_acc: 0.7362 - val_loss: 1.7657

Epoch 9/10

625/625  **3s** 5ms/step - acc: 0.9966 - loss: 0.0105 - val_acc: 0.7558 - val_loss: 1.6739

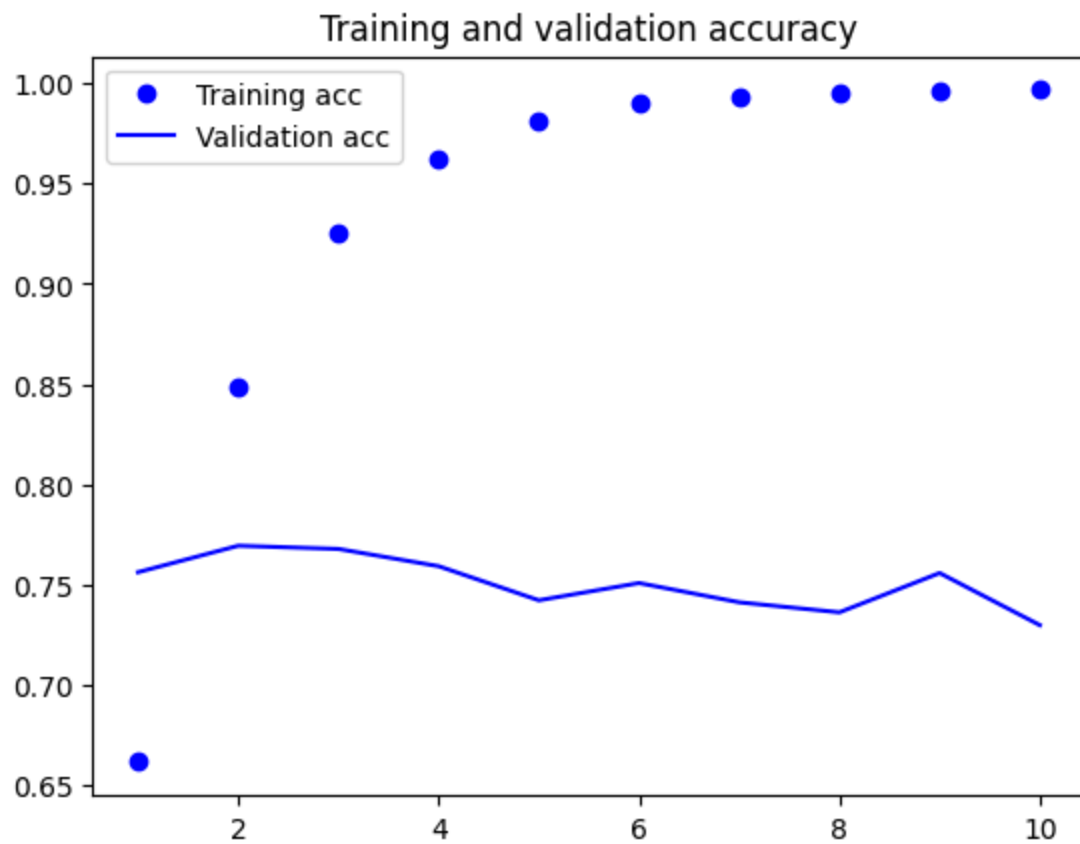
Epoch 10/10

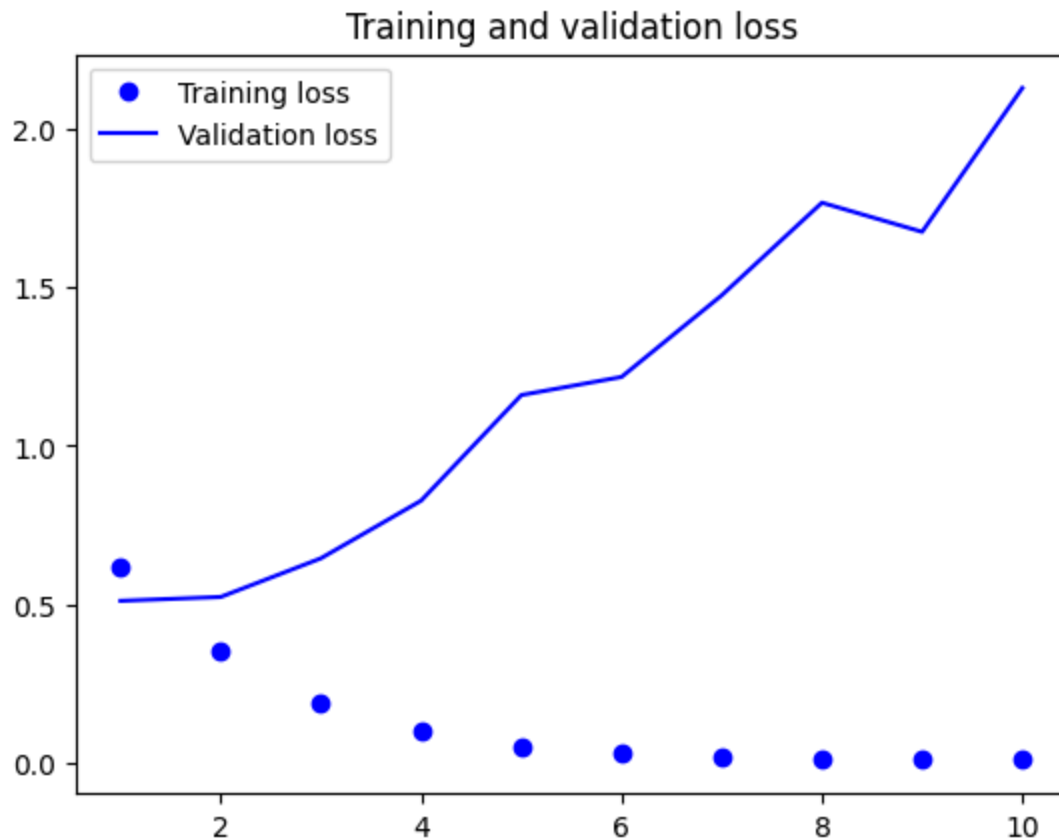
625/625  **4s** 3ms/step - acc: 0.9970 - loss: 0.0086 - val_acc: 0.7298 - val_loss: 2.1272

```
In [ ]: %matplotlib inline
import matplotlib.pyplot as plt
def plot_history(history):
    acc = history.history['acc']
    val_acc = history.history['val_acc']
    loss = history.history['loss']
    val_loss = history.history['val_loss']
    epochs = range(1, len(acc) + 1)
    plt.plot(epochs, acc, 'bo', label='Training acc')
    plt.plot(epochs, val_acc, 'b', label='Validation acc')
    plt.title('Training and validation accuracy')
    plt.legend()
    plt.figure()
    plt.plot(epochs, loss, 'bo', label='Training loss')
```

```
plt.plot(epochs, val_loss, 'b', label='Validation loss')  
plt.title('Training and validation loss')  
plt.legend()  
plt.show()
```

```
In [ ]: plot_history(history)
```





8. Fine-Tuned Dense Model

Unfreeze the embedding layer and retrain jointly with the dense head.

```
In [ ]: from keras.models import Sequential
from keras.layers import Embedding, Flatten, Dense
assert embedding_matrix.shape == (max_words, embedding_dim), "embedding_matrix shape mismatch"
model = Sequential()
model.add(Embedding(input_dim=max_words,
                    output_dim=embedding_dim,
                    input_length=maxlen,
                    weights=[embedding_matrix],
                    trainable=True))
model.add(Flatten())
model.add(Dense(32, activation='relu'))
model.add(Dense(1, activation='sigmoid'))
model.compile(optimizer='rmsprop',
              loss='binary_crossentropy',
              metrics=['acc'])
history = model.fit(x_train, y_train,
                   epochs=10,
                   batch_size=32,
                   validation_data=(x_val, y_val))
```

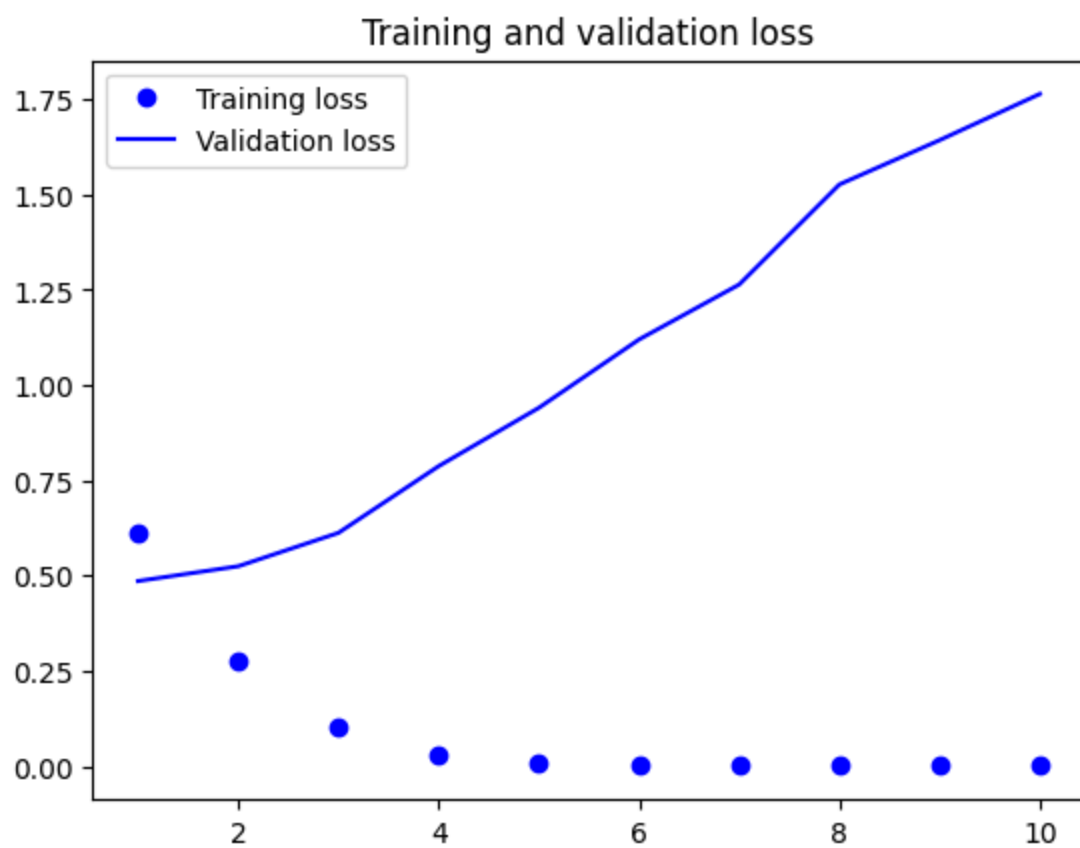
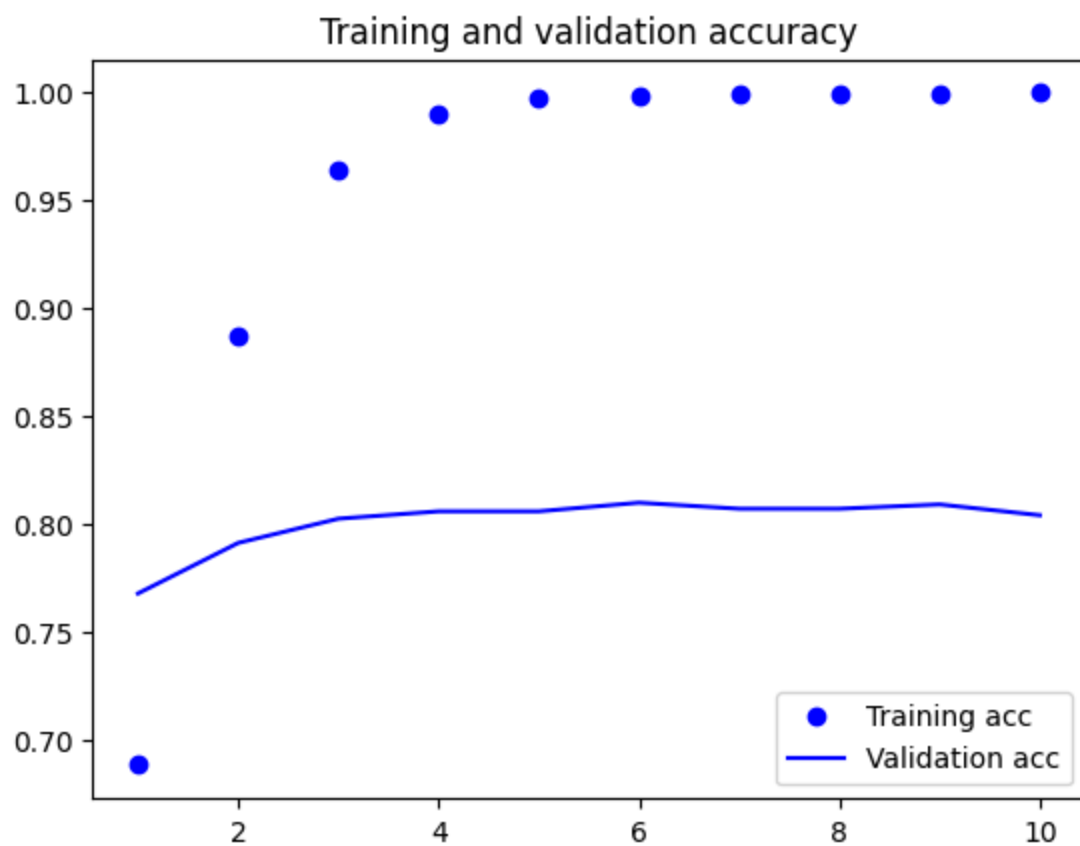
Epoch 1/10

```
/usr/local/lib/python3.12/dist-packages/keras/src/layers/core/embedding.py:9
7: UserWarning: Argument `input_length` is deprecated. Just remove it.
warnings.warn(
```

625/625  **33s** 45ms/step - acc: 0.6134 - loss: 0.8336 - val_acc: 0.7680 - val_loss: 0.4857
Epoch 2/10
625/625  **28s** 44ms/step - acc: 0.8846 - loss: 0.2872 - val_acc: 0.7914 - val_loss: 0.5245
Epoch 3/10
625/625  **27s** 43ms/step - acc: 0.9687 - loss: 0.0973 - val_acc: 0.8026 - val_loss: 0.6122
Epoch 4/10
625/625  **25s** 40ms/step - acc: 0.9925 - loss: 0.0275 - val_acc: 0.8060 - val_loss: 0.7873
Epoch 5/10
625/625  **21s** 34ms/step - acc: 0.9972 - loss: 0.0101 - val_acc: 0.8060 - val_loss: 0.9404
Epoch 6/10
625/625  **18s** 29ms/step - acc: 0.9986 - loss: 0.0039 - val_acc: 0.8100 - val_loss: 1.1198
Epoch 7/10
625/625  **17s** 26ms/step - acc: 0.9988 - loss: 0.0022 - val_acc: 0.8072 - val_loss: 1.2641
Epoch 8/10
625/625  **14s** 23ms/step - acc: 0.9993 - loss: 9.7830e-04 - val_acc: 0.8072 - val_loss: 1.5268
Epoch 9/10
625/625  **13s** 20ms/step - acc: 0.9994 - loss: 5.5005e-04 - val_acc: 0.8092 - val_loss: 1.6431
Epoch 10/10
625/625  **12s** 19ms/step - acc: 0.9997 - loss: 2.1541e-04 - val_acc: 0.8042 - val_loss: 1.7643

Training curves for the dense model.

```
In [ ]: plot_history(history)
```



```
In [ ]: embedding_matrix1 = model.layers[0].get_weights()[0]
```


9. LSTM Model

Replace Flatten+Dense with an LSTM layer to capture word order in reviews.

```
In [ ]: from keras.models import Sequential
        from keras.layers import Embedding, LSTM, Dense
        assert embedding_matrix.shape == (max_words, embedding_dim), "embedding_matrix shape mismatch"
        model = Sequential()
        model.add(Embedding(input_dim=max_words,
                             output_dim=embedding_dim,
                             input_length=maxlen,
                             weights=[embedding_matrix],
                             trainable=True))
        model.add(LSTM(32))
        model.add(Dense(1, activation='sigmoid'))
        model.compile(optimizer='rmsprop',
                      loss='binary_crossentropy',
                      metrics=['acc'])
        history = model.fit(x_train, y_train,
                             epochs=10,
                             batch_size=32,
                             validation_data=(x_val, y_val))
```

Epoch 1/10

625/625 ————— 17s 22ms/step - acc: 0.6980 - loss: 0.5643 - val_acc: 0.8540 - val_loss: 0.3514

Epoch 2/10

625/625 ————— 14s 22ms/step - acc: 0.8659 - loss: 0.3304 - val_acc: 0.8850 - val_loss: 0.2834

Epoch 3/10

625/625 ————— 14s 22ms/step - acc: 0.9106 - loss: 0.2302 - val_acc: 0.8908 - val_loss: 0.2726

Epoch 4/10

625/625 ————— 13s 22ms/step - acc: 0.9327 - loss: 0.1780 - val_acc: 0.8924 - val_loss: 0.2725

Epoch 5/10

625/625 ————— 13s 21ms/step - acc: 0.9545 - loss: 0.1323 - val_acc: 0.8960 - val_loss: 0.2968

Epoch 6/10

625/625 ————— 13s 21ms/step - acc: 0.9686 - loss: 0.0966 - val_acc: 0.8958 - val_loss: 0.3159

Epoch 7/10

625/625 ————— 13s 21ms/step - acc: 0.9786 - loss: 0.0716 - val_acc: 0.8918 - val_loss: 0.3688

Epoch 8/10

625/625 ————— 13s 21ms/step - acc: 0.9866 - loss: 0.0472 - val_acc: 0.8774 - val_loss: 0.4744

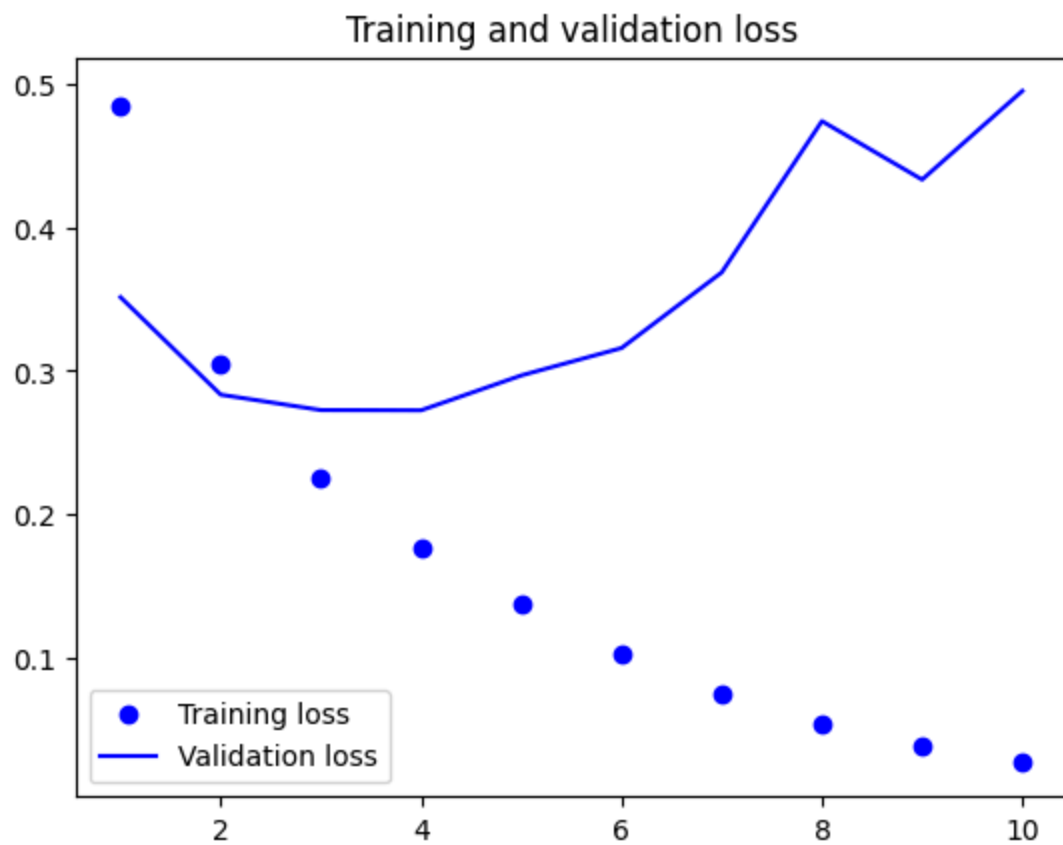
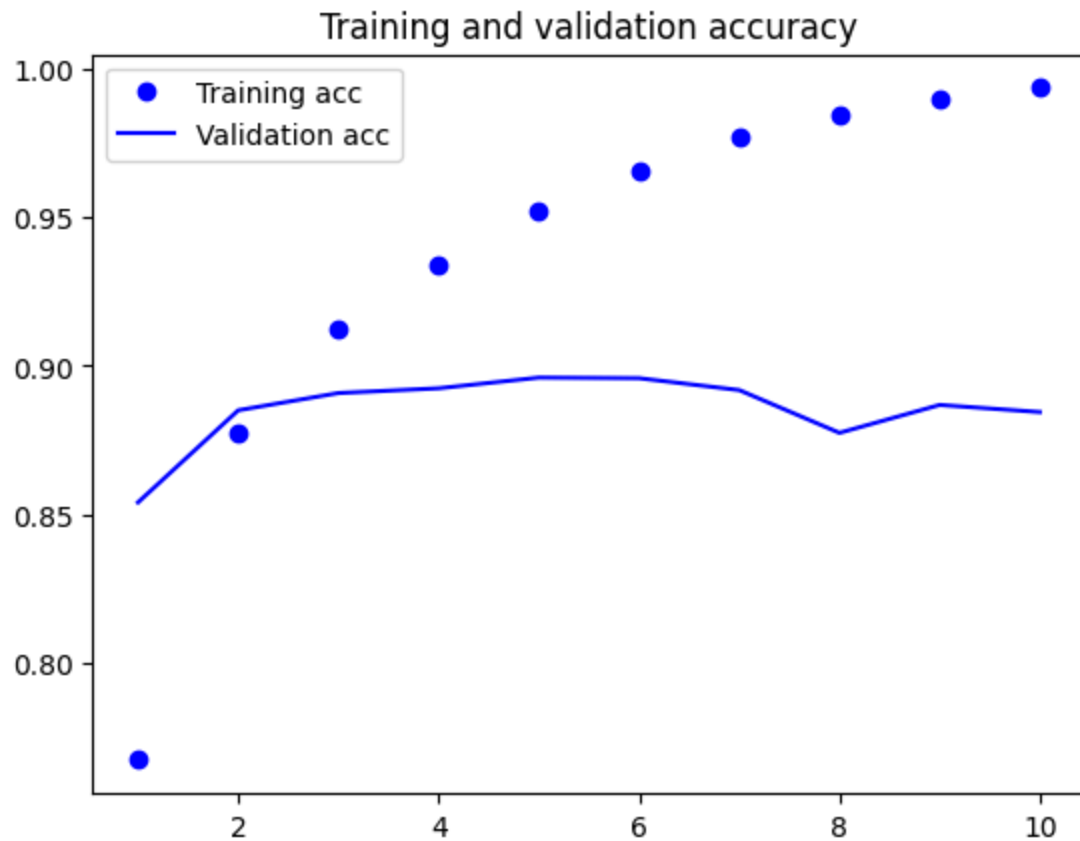
Epoch 9/10

625/625 ————— 14s 22ms/step - acc: 0.9900 - loss: 0.0383 - val_acc: 0.8868 - val_loss: 0.4336

Epoch 10/10

625/625 ————— 14s 22ms/step - acc: 0.9937 - loss: 0.0250 - val_acc: 0.8844 - val_loss: 0.4955

```
In [ ]: plot_history(history)
```



```
In [ ]: embedding_matrix2 = model.layers[0].get_weights()[0]
```

```
In [ ]: embedding_matrix2.shape
```

```
Out [ ]: (10000, 300)
```

10. Embedding Comparison — Word Distance

Cosine distance between word pairs across GloVe, dense-tuned, and LSTM-tuned embeddings.

```
In [ ]: def word_dist(word1, word2, embedding_matrix):
        if (word1 not in word_index) or (word2 not in word_index):
            return None
        idx1, idx2 = word_index[word1], word_index[word2]
        if idx1 >= max_words or idx2 >= max_words:
            return None
        vec1 = embedding_matrix[idx1]
        vec2 = embedding_matrix[idx2]
        num = np.dot(vec1, vec2)
        denom = np.linalg.norm(vec1) * np.linalg.norm(vec2)
        if denom == 0:
            return None
        cosine_sim = num / denom
        cosine_dist = 1 - cosine_sim
        return cosine_dist
    def dist_glove(word1, word2):
        return word_dist(word1, word2, embedding_matrix)
    def dist_dense(word1, word2):
        return word_dist(word1, word2, embedding_matrix1)
    def dist_LSTM(word1, word2):
        return word_dist(word1, word2, embedding_matrix2)
```

Compare whether fine-tuned embeddings place antonyms farther apart than unrelated word pairs.

```
In [ ]: print(dist_glove('better', 'worse') < dist_glove('worst', 'worse'))
        print(dist_dense('better', 'worse') < dist_dense('worst', 'worse'))
        print(dist_LSTM('better', 'worse') < dist_LSTM('worst', 'worse'))
```

```
True
False
False
```

```
In [ ]: print(dist_glove('good', 'bad') < dist_glove('poor', 'bad'))
        print(dist_dense('good', 'bad') < dist_dense('poor', 'bad'))
        print(dist_LSTM('good', 'bad') < dist_LSTM('poor', 'bad'))
```

```
True
False
True
```

```
In [ ]: print(dist_glove('bad', 'good') < dist_glove('great', 'good'))
        print(dist_dense('bad', 'good') < dist_dense('great', 'good'))
        print(dist_LSTM('bad', 'good') < dist_LSTM('great', 'good'))
```

```
True
False
False
```

```
In [ ]: print(dist_glove('clever', 'clumsy') < dist_glove('clever', 'ingenious'))
        print(dist_dense('clever', 'clumsy') < dist_dense('clever', 'ingenious'))
        print(dist_LSTM('clever', 'clumsy') < dist_LSTM('clever', 'ingenious'))
```

```
False
False
False
```

11. Nearest-Neighbor Analysis

For each embedding space, find the closest vocabulary word by cosine similarity.

```
In [ ]: def NN_general(word, embedding_matrix):
        if word not in word_index:
            return None
        idx = word_index[word]
        if idx >= max_words:
            return None
        target_vec = embedding_matrix[idx]
        normed = embedding_matrix / np.linalg.norm(embedding_matrix, axis=1, keepdims=True)
        target_norm = target_vec / np.linalg.norm(target_vec)
        sims = np.dot(normed, target_norm)
        sims[idx] = -1
        nearest_idx = np.argmax(sims)
        inv_index = {v: k for k, v in word_index.items() if v < max_words}
        nearest_word = inv_index.get(nearest_idx, None)
        return nearest_word
    def NN_LSTM(word):
        return NN_general(word, embedding_matrix2)
    def NN_dense(word):
        return NN_general(word, embedding_matrix1)
    def NN_glove(word):
        return NN_general(word, embedding_matrix)
```

```
In [ ]: EPS = 1e-10
        max_idx = max_words
        inv_index = np.array([None]*max_idx, dtype=object)
        for w, i in word_index.items():
            if i < max_idx:
                inv_index[i] = w
        def precompute_normed(M):
            norms = np.linalg.norm(M, axis=1, keepdims=True)
            norms[norms == 0] = 1.0
            N = M / norms
            mask = (np.linalg.norm(M, axis=1) > EPS)
            return N, mask
        norm_glove, mask_glove = precompute_normed(embedding_matrix)
        norm_dense, mask_dense = precompute_normed(embedding_matrix1)
        norm_lstm, mask_lstm = precompute_normed(embedding_matrix2)
        def NN_general(word, normed, mask):
```

```

idx = word_index.get(word)
if idx is None or idx >= max_idx or not mask[idx]:
    return None
if word not in word_index or word_index[word] >= max_words:
    return None
sims = normed @ normed[idx]
sims[~mask] = -1e9
sims[0] = -1e9
sims[idx] = -1e9
j = int(np.argmax(sims))
return inv_index[j]
def NN_glove(w): return NN_general(w, norm_glove, mask_glove)
def NN_dense(w): return NN_general(w, norm_dense, mask_dense)
def NN_LSTM(w): return NN_general(w, norm_lstm, mask_lstm)

```

Words whose LSTM nearest neighbor differs from GloVe but matches dense.

```

In [ ]: count = 0
for word in L_POS + L_NEG:
    if NN_LSTM(word) != NN_glove(word) and NN_dense(word) == NN_glove(word):
        print(':::', word, NN_LSTM(word), NN_glove(word))
        count += 1
print(count, 'changes', 'out of', len(L_POS) + len(L_NEG))

```

:::: admirer admiration avid
:::: awe inspiring disbelief
:::: bonus extra cd
:::: catchy snappy tunes
:::: credible plausible reliable
:::: decency morality fairness
:::: delightful lovely charming
:::: fantastically wonderfully unbelievably
:::: fine well art
:::: fresh bring amid
:::: hardy winters bloom
:::: healthy eating stable
:::: helped help helping
:::: honest truthful straightforward
:::: improve improved enhance
:::: loyal followers faithful
:::: luck lucky good
:::: mighty beast big
:::: modest substantial relatively
:::: notably including particularly
:::: openly gays criticize
:::: peaceful peace calm
:::: pleasantly delightfully amazed
:::: promised promises promise
:::: restored converted built
:::: right back left
:::: suave dashing seductive
:::: valuable useful precious
:::: wonderfully delightfully beautifully
:::: alarm alert warning
:::: annoyingly amusingly cliched
:::: assassin murderer killer
:::: bash celebration birthday
:::: bland dull boring
:::: brainless moronic idiotic
:::: cannibal zombies zombie
:::: confused frightened confusing
:::: confusing complicated confused
:::: dim bright faint
:::: drags slows dragged
:::: dreary drab gloomy
:::: fiasco botched disastrous
:::: flair charisma wit
:::: flawed flaws misguided
:::: fright wig horror
:::: fuzzy logic cute
:::: havoc causing mayhem
:::: heartbreaking poignant shocking
:::: ignore ignored ignoring
:::: illusion perception impression
:::: insane lunatic clown
:::: ironically though perhaps
:::: kill kills killing
:::: lifeless limp corpse
:::: limited only available
:::: loneliness despair boredom

```
:::: madman psychopath maniac
:::: midget hockey wrestler
:::: mistakes errors mistake
:::: perverted perverse misguided
:::: rejects reject rejected
:::: ridiculously unbelievably outrageously
:::: sarcastic condescending ironic
:::: shadowy underworld murky
:::: simplistic cliched formulaic
:::: sorely desperately lacking
:::: surrender refused accept
:::: trashy campy raunchy
:::: unemployed jobs worker
:::: unhappy frustrated disappointed
:::: unknown exact unclear
:::: unpleasant irritating unsettling
:::: whiny obnoxious whining
73 changes out of 1870
```

Words whose dense nearest neighbor differs from GloVe but matches LSTM.

```
In [ ]: count = 0
        for word in L_POS + L_NEG:
            if NN_dense(word) != NN_glove(word) and NN_LSTM(word) == NN_glove(word):
                print('::::', word, NN_dense(word), NN_glove(word))
                count += 1
        print(count, 'changes', 'out of', len(L_POS) + len(L_NEG))
```

```

:::: better even good
:::: encourage encouraged promote
:::: fantastic amazing incredible
:::: fun enjoyable stuff
:::: gorgeous lovely beautiful
:::: nice lovely wonderful
:::: perfect wonderful ideal
:::: perfection simplicity perfect
:::: poignant hilarious haunting
:::: satisfy fulfill desires
:::: sweet flavor delicious
:::: wonderful amazing terrific
:::: anxious worried eager
:::: boring dull tedious
:::: confusion anxiety chaos
:::: cruel sadistic brutal
:::: doomed unfortunately fail
:::: drain dry pour
:::: evil villain wicked
:::: fear fears worry
:::: haunting poignant eerie
:::: ignorant stupid naive
:::: kills destroys killed
:::: liar cheat thief
:::: mess awful messy
:::: slow slower pace
:::: stricken disaster survivors
:::: toxic wastes waste
:::: trouble difficulty problems
29 changes out of 1870

```

```

In [ ]: count = 0
        for word in L_POS + L_NEG:
            if NN_LSTM(word) != NN_dense(word) and NN_dense(word) != NN_glove(word):
                print('::::', word, NN_glove(word), NN_dense(word), NN_LSTM(word))
                count += 1
        print(count)

```

```

:::: angel gabriel jose lopez
:::: good better always well
:::: crap stuff woman's freaking
:::: poor poverty bad lack
:::: wrong think something why
5

```

Words where dense and LSTM agree on nearest neighbor but GloVe disagrees.

```

In [ ]: count = 0
        for word in L_POS + L_NEG:
            if NN_LSTM(word) == NN_dense(word) and NN_dense(word) != NN_glove(word):
                print('::::', word, NN_glove(word), NN_dense(word), NN_LSTM(word))
                count += 1
        print(count)

```



```

:::: best better well well
:::: clear yet clearly clearly
:::: convenient easy easier easier
:::: decent good respectable respectable
:::: educated college studied studied
:::: enough so needed needed
:::: excellent good superb superb
:::: extraordinarily incredibly exceedingly exceedingly
:::: fascinating interesting intriguing intriguing
:::: flashy stylish flamboyant flamboyant
:::: free freedom without without
:::: friendly ties relations relations
:::: genius geniuses brilliant brilliant
:::: kudos deserves acclaim acclaim
:::: outstanding achievement exceptional exceptional
:::: powerful potent strong strong
:::: realistic unrealistic realistically realistically
:::: solid consistent strong strong
:::: stable healthy relatively relatively
:::: superb brilliant excellent excellent
:::: winning won win win
:::: abomination unforgivable monstrosity monstrosity
:::: annoyance embarrassment frustration frustration
:::: bad good worse worse
:::: bored tired boring boring
:::: comical hilarious amusing amusing
:::: died death killed killed
:::: hopelessly hopeless woefully woefully
:::: inevitably inevitable surely surely
:::: isolated areas remote remote
:::: itchy skin irritating irritating
:::: jam hop band band
:::: lackluster dismal disappointing disappointing
:::: loathing paranoia disdain disdain
:::: lousy crummy awful awful
:::: overlook overlooked ignore ignore
:::: pointless meaningless useless useless
:::: repulsive disgusting unattractive unattractive
:::: slack picking weak weak
:::: sleazy seedy smarmy smarmy
:::: suck sucked sucks sucks
:::: taboo homosexuality topic topic
:::: terribly incredibly horribly horribly
:::: unfortunately fortunately nothing nothing
:::: wasted chances wasting wasting
:::: worried worry concerned concerned
:::: worse even bad bad

```

47

Words where all three embedding spaces agree on nearest neighbor.

```

In [ ]: count = 0
        for word in L_POS + L_NEG:
            if NN_LSTM(word) == NN_dense(word) and NN_dense(word) == NN_glove(word):
                print('::::', word, NN_glove(word), NN_dense(word), NN_LSTM(word))

```

```
count += 1  
print(count)
```

:::: abound plenty plenty plenty
:::: abundance diversity diversity diversity
:::: accessible access access access
:::: acclaim garnered garnered garnered
:::: acclaimed renowned renowned renowned
:::: accomplish achieve achieve achieve
:::: accomplished accomplish accomplish accomplish
:::: accomplishment achievement achievement achievement
:::: accurate precise precise precise
:::: accurately correctly correctly correctly
:::: achievement achievements achievements achievements
:::: achievements achievement achievement achievement
:::: adequate sufficient sufficient sufficient
:::: admirable commendable commendable commendable
:::: admirably excellently excellently excellently
:::: admiration affection affection affection
:::: admire appreciate appreciate appreciate
:::: adorable cute cute cute
:::: adore despise despise despise
:::: advanced technology technology technology
:::: advantage opportunity opportunity opportunity
:::: adventurous inventive inventive inventive
:::: affection admiration admiration admiration
:::: afford pay pay pay
:::: amazed thrilled thrilled thrilled
:::: amazing incredible incredible incredible
:::: amazingly astonishingly astonishingly astonishingly
:::: ambitious plan plan plan
:::: ample sufficient sufficient sufficient
:::: amuse entertain entertain entertain
:::: amusing hilarious hilarious hilarious
:::: amusingly hilariously hilariously hilariously
:::: appeal appeals appeals appeals
:::: appealing appeal appeal appeal
:::: applaud admire admire admire
:::: appreciate appreciated appreciated appreciated
:::: appreciated appreciate appreciate appreciate
:::: appropriate necessary necessary necessary
:::: aspirations ambitions ambitions ambitions
:::: assure ensure ensure ensure
:::: astonishing astounding astounding astounding
:::: astonishingly amazingly amazingly amazingly
:::: astounding astonishing astonishing astonishing
:::: attraction tourist tourist tourist
:::: attractive unattractive unattractive unattractive
:::: authentic authenticity authenticity authenticity
:::: available provide provide provide
:::: avid enthusiastic enthusiastic enthusiastic
:::: award awards awards awards
:::: awarded award award award
:::: awards award award award
:::: awesome amazing amazing amazing
:::: balanced balance balance balance
:::: bargain buying buying buying
:::: beautiful lovely lovely lovely
:::: beautifully wonderfully wonderfully wonderfully

:::: beauty beautiful beautiful beautiful
:::: believable likable likable likable
:::: beloved loved loved loved
:::: benefit benefits benefits benefits
:::: benefits benefit benefit benefit
:::: bless god god god
:::: bliss insanity insanity insanity
:::: blockbuster blockbusters blockbusters blockbusters
:::: bloom flowers flowers flowers
:::: boom bubble bubble bubble
:::: boost increase increase increase
:::: brave courageous courageous courageous
:::: bravery courage courage courage
:::: bravo lara lara lara
:::: breakthrough success success success
:::: breathtaking stunning stunning stunning
:::: bright colors colors colors
:::: brilliance flashes flashes flashes
:::: brilliant superb superb superb
:::: brilliantly superbly superbly superbly
:::: calm quiet quiet quiet
:::: capable incapable incapable incapable
:::: captivating mesmerizing mesmerizing mesmerizing
:::: celebrate celebrating celebrating celebrating
:::: celebrated celebrating celebrating celebrating
:::: celebration celebrating celebrating celebrating
:::: champion defending defending defending
:::: charisma charm charm charm
:::: charismatic flamboyant flamboyant flamboyant
:::: charm charms charms charms
:::: charming lovely lovely lovely
:::: cheaper cheap cheap cheap
:::: cheer cheers cheers cheers
:::: cheerful playful playful playful
:::: clarity simplicity simplicity simplicity
:::: classic classics classics classics
:::: classy stylish stylish stylish
:::: clean cleaning cleaning cleaning
:::: clearly obviously obviously obviously
:::: clever ingenious ingenious ingenious
:::: cleverly expertly expertly expertly
:::: coherent cohesive cohesive cohesive
:::: cohesive coherent coherent coherent
:::: colorful colourful colourful colourful
:::: comfort warmth warmth warmth
:::: comfortable relaxed relaxed relaxed
:::: commendable admirable admirable admirable
:::: commitment determination determination determination
:::: compassion empathy empathy empathy
:::: compassionate compassion compassion compassion
:::: compliment insult insult insult
:::: confidence optimism optimism optimism
:::: confident optimistic optimistic optimistic
:::: consistent inconsistent inconsistent inconsistent
:::: consistently repeatedly repeatedly repeatedly
:::: continuity semblance semblance semblance

:::: contribution contribute contribute contribute
:::: conveniently forgets forgets forgets
:::: convincing compelling compelling compelling
:::: convincingly convincing convincing convincing
:::: cool warm warm warm
:::: coolest funniest funniest funniest
:::: correct incorrect incorrect incorrect
:::: correctly accurately accurately accurately
:::: courage bravery bravery bravery
:::: courageous brave brave brave
:::: creative creativity creativity creativity
:::: crisp bright bright bright
:::: cure treat treat treat
:::: cute adorable adorable adorable
:::: daring bold bold bold
:::: darling cameron cameron cameron
:::: dashing suave suave suave
:::: dawn midnight midnight midnight
:::: dazzling brilliant brilliant brilliant
:::: dedicated devoted devoted devoted
:::: defeat victory victory victory
:::: defeated defeat defeat defeat
:::: delicate fragile fragile fragile
:::: delicious tasty tasty tasty
:::: delight joy joy joy
:::: delighted thrilled thrilled thrilled
:::: delightfully deliciously deliciously deliciously
:::: deservedly deserved deserved deserved
:::: deserving worthy worthy worthy
:::: destiny fate fate fate
:::: dignity respect respect respect
:::: distinction distinguished distinguished distinguished
:::: distinctive characteristic characteristic characteristic
:::: distinguished distinction distinction distinction
:::: divine god god god
:::: dominate dominated dominated dominated
:::: dominated dominate dominate dominate
:::: dynamic static static static
:::: eager hoping hoping hoping
:::: eagerly waited waited waited
:::: earnest sincere sincere sincere
:::: ease tensions tensions tensions
:::: easier harder harder harder
:::: easy easier easier easier
:::: ecstasy heroin heroin heroin
:::: effective efficient efficient efficient
:::: effectively essentially essentially essentially
:::: efficient effective effective effective
:::: effortlessly expertly expertly expertly
:::: elegant stylish stylish stylish
:::: elevate enhance enhance enhance
:::: elite ranks ranks ranks
:::: empathize sympathize sympathize sympathize
:::: empathy compassion compassion compassion
:::: enchanted enchanting enchanting enchanting
:::: enchanting captivating captivating captivating

:::: endearing likable likable likable
:::: energetic enthusiastic enthusiastic enthusiastic
:::: engaging engage engage engage
:::: engrossing enthralling enthralling enthralling
:::: enhance improve improve improve
:::: enhanced enhance enhance enhance
:::: enhances enhance enhance enhance
:::: enjoy enjoying enjoying enjoying
:::: enjoyable entertaining entertaining entertaining
:::: enjoyed enjoying enjoying enjoying
:::: enjoying enjoyed enjoyed enjoyed
:::: enjoyment pleasure pleasure pleasure
:::: enjoys enjoyed enjoyed enjoyed
:::: entertain entertained entertained entertained
:::: entertaining enjoyable enjoyable enjoyable
:::: entertains entertain entertain entertain
:::: enthusiasm excitement excitement excitement
:::: enthusiastic passionate passionate passionate
:::: envy jealousy jealousy jealousy
:::: exceedingly extraordinarily extraordinarily extraordinarily
:::: excellence achievement achievement achievement
:::: excellently superbly superbly superbly
:::: excels concentrates concentrates concentrates
:::: exceptional extraordinary extraordinary extraordinary
:::: exceptionally extraordinarily extraordinarily extraordinarily
:::: excited thrilled thrilled thrilled
:::: excitement enthusiasm enthusiasm enthusiasm
:::: exciting interesting interesting interesting
:::: expertly cleverly cleverly cleverly
:::: exquisite beautifully beautifully beautifully
:::: extraordinary exceptional exceptional exceptional
:::: fabulous wonderful wonderful wonderful
:::: fair reasonable reasonable reasonable
:::: fairly quite quite quite
:::: fairness honesty honesty honesty
:::: faith belief belief belief
:::: faithful loyal loyal loyal
:::: fame hall hall hall
:::: famed famous famous famous
:::: famous famed famed famed
:::: fancy flashy flashy flashy
:::: fans crowd crowd crowd
:::: fascination obsession obsession obsession
:::: fast slow slow slow
:::: faster slower slower slower
:::: favor favour favour favour
:::: favorite favorites favorites favorites
:::: favour favor favor favor
:::: feat accomplish accomplish accomplish
:::: fiery temper temper temper
:::: finely chopped chopped chopped
:::: finest greatest greatest greatest
:::: flawless impeccable impeccable impeccable
:::: fond memories memories memories
:::: foremost greatest greatest greatest
:::: fortunate lucky lucky lucky

:::: fortunately luckily luckily luckily
:::: fortune wealth wealth wealth
:::: freed imprisoned imprisoned imprisoned
:::: freedom liberty liberty liberty
:::: futuristic retro retro retro
:::: gain gaining gaining gaining
:::: gained gaining gaining gaining
:::: gaining gained gained gained
:::: gem gems gems gems
:::: gems gem gem gem
:::: generous benefits benefits benefits
:::: gentle playful playful playful
:::: genuine sincere sincere sincere
:::: gifted talented talented talented
:::: glad pleased pleased pleased
:::: gladly willingly willingly willingly
:::: glamorous sexy sexy sexy
:::: glee joy joy joy
:::: glorious splendid splendid splendid
:::: glory greatness greatness greatness
:::: glow glowing glowing glowing
:::: glowing glow glow glow
:::: gold silver silver silver
:::: golden awards awards awards
:::: goodness kindness kindness kindness
:::: grace mary mary mary
:::: grand slam slam slam
:::: grateful thankful thankful thankful
:::: great greatest greatest greatest
:::: greatest great great great
:::: greatness glory glory glory
:::: grin smile smile smile
:::: groundbreaking landmark landmark landmark
:::: guarantee guaranteed guaranteed guaranteed
:::: gusto subtlety subtlety subtlety
:::: hale irwin irwin irwin
:::: hallmark signature signature signature
:::: handsome charming charming charming
:::: handy useful useful useful
:::: happier happy happy happy
:::: happily quietly quietly quietly
:::: happiness joy joy joy
:::: happy glad glad glad
:::: harmless useless useless useless
:::: harmony melody melody melody
:::: heartfelt sincere sincere sincere
:::: heartwarming poignant poignant poignant
:::: heaven hell hell hell
:::: heavenly heaven heaven heaven
:::: helpful useful useful useful
:::: helping help help help
:::: hero heroes heroes heroes
:::: heroic courageous courageous courageous
:::: heroine protagonist protagonist protagonist
:::: hilarious amusing amusing amusing
:::: holy sacred sacred sacred

:::: homage tribute tribute tribute
:::: honesty sincerity sincerity sincerity
:::: honor honour honour honour
:::: honorable courageous courageous courageous
:::: hopeful optimistic optimistic optimistic
:::: hot cool cool cool
:::: hug goodbye goodbye goodbye
:::: humble upbringing upbringing upbringing
:::: humor humour humour humour
:::: humorous amusing amusing amusing
:::: humour humor humor humor
:::: ideal suitable suitable suitable
:::: idol contestant contestant contestant
:::: imaginative inventive inventive inventive
:::: immense enormous enormous enormous
:::: impeccable flawless flawless flawless
:::: important crucial crucial crucial
:::: impress convince convince convince
:::: impressed surprised surprised surprised
:::: impressive remarkable remarkable remarkable
:::: improved improve improve improve
:::: improvement improved improved improved
:::: incredible amazing amazing amazing
:::: incredibly extraordinarily extraordinarily extraordinarily
:::: indulgent overbearing overbearing overbearing
:::: influential prominent prominent prominent
:::: ingenious inventive inventive inventive
:::: innovative inventive inventive inventive
:::: insightful thoughtful thoughtful thoughtful
:::: inspiration inspired inspired inspired
:::: inspirational inspiring inspiring inspiring
:::: inspire inspiring inspiring inspiring
:::: inspiring inspire inspire inspire
:::: instantly suddenly suddenly suddenly
:::: integral function function function
:::: integrated systems systems systems
:::: intelligence cia cia cia
:::: intelligent smart smart smart
:::: interesting intriguing intriguing intriguing
:::: interests business business business
:::: intimate relationships relationships relationships
:::: intricate elaborate elaborate elaborate
:::: intrigue suspense suspense suspense
:::: intriguing interesting interesting interesting
:::: inventive imaginative imaginative imaginative
:::: irresistible seductive seductive seductive
:::: jolly merry merry merry
:::: joy delight delight delight
:::: keen eager eager eager
:::: kindly gentle gentle gentle
:::: kindness compassion compassion compassion
:::: lavish gifts gifts gifts
:::: lead leads leads leads
:::: leading led led led
:::: leads lead lead lead
:::: lean muscular muscular muscular

:::: led leading leading leading
:::: legendary legend legend legend
:::: liberty freedom freedom freedom
:::: lighter thinner thinner thinner
:::: likable likeable likeable likeable
:::: like such such such
:::: liked loved loved loved
:::: likes loves loves loves
:::: liking dislike dislike dislike
:::: lively vibrant vibrant vibrant
:::: logical rational rational rational
:::: lovable likable likable likable
:::: love loves loves loves
:::: loved liked liked liked
:::: lovely beautiful beautiful beautiful
:::: lover boyfriend boyfriend boyfriend
:::: loves hates hates hates
:::: loving caring caring caring
:::: loyalty devotion devotion devotion
:::: lucky fortunate fortunate fortunate
:::: lush gorgeous gorgeous gorgeous
:::: luxury hotel hotel hotel
:::: lyrical poetic poetic poetic
:::: magic magical magical magical
:::: magical magic magic magic
:::: magnificent splendid splendid splendid
:::: marvel comics comics comics
:::: marvellous marvelous marvelous marvelous
:::: marvelous wonderful wonderful wonderful
:::: master bachelor bachelor bachelor
:::: masterful superb superb superb
:::: masterfully expertly expertly expertly
:::: masterpiece masterpieces masterpieces masterpieces
:::: masterpieces masterpiece masterpiece masterpiece
:::: masters master master master
:::: mature immature immature immature
:::: maturity mature mature mature
:::: meaningful useful useful useful
:::: memorable unforgettable unforgettable unforgettable
:::: mercifully thankfully thankfully thankfully
:::: mercy compassion compassion compassion
:::: merit awarded awarded awarded
:::: merry jolly jolly jolly
:::: mesmerizing captivating captivating captivating
:::: miracle cure cure cure
:::: miraculously magically magically magically
:::: modern contemporary contemporary contemporary
:::: modesty honesty honesty honesty
:::: monumental gigantic gigantic gigantic
:::: morality morals morals morals
:::: motivated politically politically politically
:::: neat neatly neatly neatly
:::: neatly neat neat neat
:::: nicely beautifully beautifully beautifully
:::: nifty clever clever clever
:::: noble barnes barnes barnes

:::: noteworthy notable notable notable
:::: novelty item item item
:::: obsession fascination fascination fascination
:::: optimism enthusiasm enthusiasm enthusiasm
:::: optimistic confident confident confident
:::: originality creativity creativity creativity
:::: paradise heaven heaven heaven
:::: paramount warner warner warner
:::: passion love love love
:::: passionate enthusiastic enthusiastic enthusiastic
:::: patience restraint restraint restraint
:::: patient patients patients patients
:::: patriotic revolutionary revolutionary revolutionary
:::: peace talks talks talks
:::: perfectly beautifully beautifully beautifully
:::: phenomenal incredible incredible incredible
:::: playful whimsical whimsical whimsical
:::: pleasant lovely lovely lovely
:::: pleased delighted delighted delighted
:::: pleasing satisfying satisfying satisfying
:::: pleasure enjoyment enjoyment enjoyment
:::: poetic lyrical lyrical lyrical
:::: polished shiny shiny shiny
:::: polite thoughtful thoughtful thoughtful
:::: popular popularity popularity popularity
:::: posh trendy trendy trendy
:::: positive negative negative negative
:::: positively react react react
:::: praise praised praised praised
:::: precious valuable valuable valuable
:::: precise exact exact exact
:::: precisely exactly exactly exactly
:::: prefer tend tend tend
:::: preferably chopped chopped chopped
:::: prefers prefer prefer prefer
:::: pretty quite quite quite
:::: priceless precious precious precious
:::: pride sense sense sense
:::: prize award award award
:::: profound sorrow sorrow sorrow
:::: profoundly deeply deeply deeply
:::: progress efforts efforts efforts
:::: prolific renowned renowned renowned
:::: prominent influential influential influential
:::: promise promises promises promises
:::: promises promise promise promise
:::: promising promise promise promise
:::: promptly immediately immediately immediately
:::: proper appropriate appropriate appropriate
:::: properly adequately adequately adequately
:::: pros cons cons cons
:::: protect protecting protecting protecting
:::: protection protect protect protect
:::: protective protection protection protection
:::: proud glad glad glad
:::: proves prove prove prove

:::: proving prove prove prove
:::: pure essence essence essence
:::: quaint charming charming charming
:::: qualify qualifies qualifies qualifies
:::: quiet calm calm calm
:::: rapid slow slow slow
:::: rapture satanic satanic satanic
:::: rational logical logical logical
:::: readily easily easily easily
:::: ready prepared prepared prepared
:::: reasonable reasonably reasonably reasonably
:::: reasonably fairly fairly fairly
:::: recommend recommended recommended recommended
:::: recommendation recommended recommended recommended
:::: recommended recommend recommend recommend
:::: recover recovering recovering recovering
:::: redeem redeemed redeemed redeemed
:::: redeeming redeemed redeemed redeemed
:::: redemption forgiveness forgiveness forgiveness
:::: refreshing delicious delicious delicious
:::: regal cinemas cinemas cinemas
:::: regard respect respect respect
:::: relaxed relaxing relaxing relaxing
:::: reliable accurate accurate accurate
:::: relief aid aid aid
:::: remarkable astonishing astonishing astonishing
:::: remarkably amazingly amazingly amazingly
:::: renaissance medieval medieval medieval
:::: renowned famed famed famed
:::: reputation credibility credibility credibility
:::: respect regard regard regard
:::: respectable decent decent decent
:::: revelation revelations revelations revelations
:::: revival gothic gothic gothic
:::: revive efforts efforts efforts
:::: revolutionary revolution revolution revolution
:::: reward rewarded rewarded rewarded
:::: rewarding rewarded rewarded rewarded
:::: rich wealthy wealthy wealthy
:::: richly beautifully beautifully beautifully
:::: righteous vengeful vengeful vengeful
:::: rightfully rightly rightly rightly
:::: rightly rightfully rightfully rightfully
:::: romantic romance romance romance
:::: safe secure secure secure
:::: safely safe safe safe
:::: saint st st st
:::: sane insane insane insane
:::: satisfactory satisfied satisfied satisfied
:::: satisfied pleased pleased pleased
:::: satisfying enjoyable enjoyable enjoyable
:::: seasoned veteran veteran veteran
:::: secure ensure ensure ensure
:::: sensation phenomenon phenomenon phenomenon
:::: sensational lurid lurid lurid
:::: sensible reasonable reasonable reasonable

:::: sensitive issues issues issues
:::: sexy seductive seductive seductive
:::: sharp drop drop drop
:::: shine shines shines shines
:::: shiny polished polished polished
:::: significant substantial substantial substantial
:::: silent silence silence silence
:::: simpler easier easier easier
:::: sincere heartfelt heartfelt heartfelt
:::: sincerely sincere sincere sincere
:::: sincerity honesty honesty honesty
:::: skill skills skills skills
:::: skilled talented talented talented
:::: slick flashy flashy flashy
:::: smart intelligent intelligent intelligent
:::: smarter funnier funnier funnier
:::: smile grin grin grin
:::: smiles smile smile smile
:::: smiling laughing laughing laughing
:::: smitten intrigued intrigued intrigued
:::: smooth smoothly smoothly smoothly
:::: smoothly proceed proceed proceed
:::: snappy catchy catchy catchy
:::: soft drinks drinks drinks
:::: sophisticated sophistication sophistication sophistication
:::: spectacular stunning stunning stunning
:::: spectacularly stunningly stunningly stunningly
:::: spirited lively lively lively
:::: spiritual religious religious religious
:::: splendid magnificent magnificent magnificent
:::: spontaneous reactions reactions reactions
:::: standout rookie rookie rookie
:::: steady rate rate rate
:::: stellar solid solid solid
:::: stimulating intellectually intellectually intellectually
:::: straightforward simple simple simple
:::: striking strike strike strike
:::: strikingly remarkably remarkably remarkably
:::: strong stronger stronger stronger
:::: stronger weaker weaker weaker
:::: strongest strong strong strong
:::: stunned shocked shocked shocked
:::: stunning spectacular spectacular spectacular
:::: stunningly astonishingly astonishingly astonishingly
:::: stylish elegant elegant elegant
:::: stylized cartoonish cartoonish cartoonish
:::: sublime breathtaking breathtaking breathtaking
:::: succeed succeeds succeeds succeeds
:::: succeeded succeed succeed succeed
:::: succeeds succeed succeed succeed
:::: success achieved achieved achieved
:::: successful success success success
:::: successfully effectively effectively effectively
:::: suffice presume presume presume
:::: sufficient adequate adequate adequate
:::: suitable ideal ideal ideal

:::: super bowl bowl bowl
:::: superbly brilliantly brilliantly brilliantly
:::: superior inferior inferior inferior
:::: support supported supported supported
:::: supported support support support
:::: supporting support support support
:::: supportive enthusiastic enthusiastic enthusiastic
:::: supports supported supported supported
:::: supremacy struggle struggle struggle
:::: supreme court court court
:::: surreal comical comical comical
:::: survival survive survive survive
:::: survivor survivors survivors survivors
:::: sweeping swept swept swept
:::: sweetheart girlfriend girlfriend girlfriend
:::: swift quick quick quick
:::: talent talents talents talents
:::: talented gifted gifted gifted
:::: talents talent talent talent
:::: tender crisp crisp crisp
:::: terrific wonderful wonderful wonderful
:::: thank grateful grateful grateful
:::: thankful grateful grateful grateful
:::: thinner lighter lighter lighter
:::: thoughtful insightful insightful insightful
:::: thrill thrills thrills thrills
:::: thrilled delighted delighted delighted
:::: thrilling exciting exciting exciting
:::: thrills thrill thrill thrill
:::: tolerable bearable bearable bearable
:::: top highest highest highest
:::: tops topped topped topped
:::: tough hard hard hard
:::: transparent manner manner manner
:::: treasure priceless priceless priceless
:::: tremendously enormously enormously enormously
:::: trendy fashions fashions fashions
:::: triumph victory victory victory
:::: trust mutual mutual mutual
:::: trusted respected respected respected
:::: truthful honest honest honest
:::: understandable justified justified justified
:::: unforgettable memorable memorable memorable
:::: unreal surreal surreal surreal
:::: upbeat downbeat downbeat downbeat
:::: uplifting inspiring inspiring inspiring
:::: useful helpful helpful helpful
:::: variety various various various
:::: versatile talented talented talented
:::: vibrant lively lively lively
:::: victory win win win
:::: virtue virtues virtues virtues
:::: vivid vividly vividly vividly
:::: warm cool cool cool
:::: warmth warm warm warm
:::: wealthy rich rich rich

:::: welcome welcomed welcomed welcomed
:::: well both both both
:::: wholesome uplifting uplifting uplifting
:::: willing unwilling unwilling unwilling
:::: willingly gladly gladly gladly
:::: win victory victory victory
:::: winner winners winners winners
:::: winners winner winner winner
:::: wins win win win
:::: wisdom knowledge knowledge knowledge
:::: wise foolish foolish foolish
:::: wisely intelligently intelligently intelligently
:::: witty insightful insightful insightful
:::: won winning winning winning
:::: wonder wondering wondering wondering
:::: wonders wonder wonder wonder
:::: wondrous marvelous marvelous marvelous
:::: woo kim kim kim
:::: work working working working
:::: worked working working working
:::: works work work work
:::: worth million million million
:::: worthwhile enjoyable enjoyable enjoyable
:::: worthy deserving deserving deserving
:::: wow yeah yeah yeah
:::: youthful enthusiasm enthusiasm enthusiasm
:::: abominable atrocious atrocious atrocious
:::: abrupt sudden sudden sudden
:::: abruptly suddenly suddenly suddenly
:::: absence lack lack lack
:::: absurd ridiculous ridiculous ridiculous
:::: absurdity silliness silliness silliness
:::: abuse sexual sexual sexual
:::: abused sexually sexually sexually
:::: abusive abuse abuse abuse
:::: abysmal dismal dismal dismal
:::: accidental unintentional unintentional unintentional
:::: addict addicted addicted addicted
:::: addicted addict addict addict
:::: afraid scared scared scared
:::: aggressive approach approach approach
:::: agony anguish anguish anguish
:::: alienate offend offend offend
:::: alienation loneliness loneliness loneliness
:::: ambiguity ambiguous ambiguous ambiguous
:::: ambiguous vague vague vague
:::: anger frustration frustration frustration
:::: angry furious furious furious
:::: anguish grief grief grief
:::: annoy confuse confuse confuse
:::: annoyed irritated irritated irritated
:::: annoying irritating irritating irritating
:::: antagonist villain villain villain
:::: anxiety anger anger anger
:::: apocalypse apocalyptic apocalyptic apocalyptic
:::: apocalyptic apocalypse apocalypse apocalypse

:::: appalled horrified horrified horrified
:::: appalling horrendous horrendous horrendous
:::: arrogance stupidity stupidity stupidity
:::: arrogant cocky cocky cocky
:::: ashamed embarrassed embarrassed embarrassed
:::: assault attack attack attack
:::: atrocious horrendous horrendous horrendous
:::: atrocities crimes crimes crimes
:::: atrocity atrocities atrocities atrocities
:::: attack attacks attacks attacks
:::: attacks attack attack attack
:::: avenge revenge revenge revenge
:::: awful horrible horrible horrible
:::: awfully unbelievably unbelievably unbelievably
:::: awfulness idiocy idiocy idiocy
:::: awkward uncomfortable uncomfortable uncomfortable
:::: backwoods redneck redneck redneck
:::: badly severely severely severely
:::: bait traps traps traps
:::: banal trite trite trite
:::: barbarian conan conan conan
:::: bashing bullying bullying bullying
:::: bastard ol' ol' ol'
:::: beg begging begging begging
:::: begging beg beg beg
:::: bent twisted twisted twisted
:::: betrayal betrayed betrayed betrayed
:::: betrays informs informs informs
:::: beware warn warn warn
:::: bias prejudice prejudice prejudice
:::: biased inaccurate inaccurate inaccurate
:::: bickering bitter bitter bitter
:::: bitch slut slut slut
:::: bitchy sassy sassy sassy
:::: biting bite bite bite
:::: bitter rivalry rivalry rivalry
:::: bizarre strange strange strange
:::: blackmail threats threats threats
:::: blah hmmm hmmm hmmm
:::: blame blamed blamed blamed
:::: blatant shameless shameless shameless
:::: blatantly blatant blatant blatant
:::: bleak gloomy gloomy gloomy
:::: bleed bleeding bleeding bleeding
:::: bleeding stomach stomach stomach
:::: blind deaf deaf deaf
:::: bloodshed violence violence violence
:::: bloody brutal brutal brutal
:::: blow blows blows blows
:::: blunt unusually unusually unusually
:::: bogus phony phony phony
:::: bomb bombs bombs bombs
:::: bondage fetish fetish fetish
:::: bore bearing bearing bearing
:::: boredom tedium tedium tedium
:::: bother bothered bothered bothered

:::: bothered bother bother bother
:::: bothering bothered bothered bothered
:::: bothers bothering bothering bothering
:::: brat whiny whiny whiny
:::: break breaking breaking breaking
:::: breakdown failure failure failure
:::: breaking break break break
:::: breaks break break break
:::: broke breaking breaking breaking
:::: broken broke broke broke
:::: brutal ruthless ruthless ruthless
:::: brutality brutal brutal brutal
:::: brutally tortured tortured tortured
:::: bug bugs bugs bugs
:::: bugs bug bug bug
:::: bullies bully bully bully
:::: bully bullies bullies bullies
:::: bullying bully bully bully
:::: burden enormous enormous enormous
:::: burn burning burning burning
:::: burned burnt burnt burnt
:::: burning burned burned burned
:::: burns burn burn burn
:::: bust boom boom boom
:::: butcher shop shop shop
:::: cancer breast breast breast
:::: caricature caricatures caricatures caricatures
:::: carnage bloodshed bloodshed bloodshed
:::: cartoonish stylized stylized stylized
:::: catastrophe disaster disaster disaster
:::: cave tomb tomb tomb
:::: challenging challenge challenge challenge
:::: chaos mayhem mayhem mayhem
:::: chaotic chaos chaos chaos
:::: cheap cheaper cheaper cheaper
:::: cheaply cheaper cheaper cheaper
:::: cheat cheating cheating cheating
:::: cheated cheating cheating cheating
:::: cheating cheated cheated cheated
:::: cheesy campy campy campy
:::: childish silly silly silly
:::: chill cold cold cold
:::: choke triangle triangle triangle
:::: choppy meandering meandering meandering
:::: chore tedious tedious tedious
:::: clash showdown showdown showdown
:::: cliché clichés clichés clichés
:::: clichéd trite trite trite
:::: cloud clouds clouds clouds
:::: clueless ignorant ignorant ignorant
:::: clumsy inept inept inept
:::: clunky clumsy clumsy clumsy
:::: coarse texture texture texture
:::: cocky arrogant arrogant arrogant
:::: cold warm warm warm
:::: collapse demise demise demise

:::: complain complained complained complained
:::: complained complaining complaining complaining
:::: complaining complained complained complained
:::: complains complain complain complain
:::: complaint complaints complaints complaints
:::: complaints complaint complaint complaint
:::: complex complicated complicated complicated
:::: complicated difficult difficult difficult
:::: concern concerns concerns concerns
:::: concerned worried worried worried
:::: concerns concern concern concern
:::: condemned criticized criticized criticized
:::: condescending sarcastic sarcastic sarcastic
:::: confess admit admit admit
:::: confession confess confess confess
:::: confined isolated isolated isolated
:::: conflict conflicts conflicts conflicts
:::: conflicted feelings feelings feelings
:::: conflicts conflict conflict conflict
:::: confront confronting confronting confronting
:::: confrontation showdown showdown showdown
:::: confuse distract distract distract
:::: cons pros pros pros
:::: conservative liberal liberal liberal
:::: conspiracy charges charges charges
:::: contempt disdain disdain disdain
:::: contrived melodramatic melodramatic melodramatic
:::: controversial controversy controversy controversy
:::: controversy debate debate debate
:::: convoluted complicated complicated complicated
:::: corrupt corruption corruption corruption
:::: corruption corrupt corrupt corrupt
:::: coward colonel colonel colonel
:::: cowardly despicable despicable despicable
:::: crack cracking cracking cracking
:::: cracks cracking cracking cracking
:::: crappy crummy crummy crummy
:::: crash accident accident accident
:::: crashed crash crash crash
:::: crashes crash crash crash
:::: crashing crashed crashed crashed
:::: crass shameless shameless shameless
:::: craven wes wes wes
:::: crazy stupid stupid stupid
:::: creep creeps creeps creeps
:::: creeps creep creep creep
:::: creepy spooky spooky spooky
:::: crime crimes crimes crimes
:::: criminal crimes crimes crimes
:::: cringe giggle giggle giggle
:::: crippled severely severely severely
:::: crisis turmoil turmoil turmoil
:::: critic reviewer reviewer reviewer
:::: critical crucial crucial crucial
:::: criticism criticisms criticisms criticisms
:::: criticisms criticism criticism criticism

:::: criticize criticized criticized criticized
:::: criticized praised praised praised
:::: critics criticism criticism criticism
:::: crook crooks crooks crooks
:::: crooked shady shady shady
:::: crooks thieves thieves thieves
:::: crowded busy busy busy
:::: crude oil oil oil
:::: cruelty brutality brutality brutality
:::: crummy lousy lousy lousy
:::: crush crushed crushed crushed
:::: crushed crush crush crush
:::: cry cries cries cries
:::: curse cursed cursed cursed
:::: cursed curse curse curse
:::: cynical jaded jaded jaded
:::: cynicism disdain disdain disdain
:::: damage damaged damaged damaged
:::: damaged destroyed destroyed destroyed
:::: damn darn darn darn
:::: damned damn damn damn
:::: danger threat threat threat
:::: dangerous potentially potentially potentially
:::: dark darker darker darker
:::: darker dark dark dark
:::: darkness shadows shadows shadows
:::: dead killed killed killed
:::: deadly attacks attacks attacks
:::: deaf mute mute mute
:::: death died died died
:::: debt credit credit credit
:::: deception manipulation manipulation manipulation
:::: decline rise rise rise
:::: deformed limbs limbs limbs
:::: defy defies defies defies
:::: delirious hysterical hysterical hysterical
:::: demise downfall downfall downfall
:::: demon demons demons demons
:::: demonic demon demon demon
:::: denial denying denying denying
:::: denied deny deny deny
:::: dense thick thick thick
:::: deny denying denying denying
:::: denying deny deny deny
:::: depressed weak weak weak
:::: depressing unsettling unsettling unsettling
:::: depression anxiety anxiety anxiety
:::: desert mountains mountains mountains
:::: desolate dusty dusty dusty
:::: despair anguish anguish anguish
:::: desperate desperately desperately desperately
:::: desperately desperate desperate desperate
:::: desperation despair despair despair
:::: despicable cowardly cowardly cowardly
:::: despise adore adore adore
:::: destroy destroying destroying destroying

:::: destruction weapons weapons weapons
:::: destructive devastating devastating devastating
:::: detract distract distract distract
:::: devastating disastrous disastrous disastrous
:::: devil satan satan satan
:::: devoid utterly utterly utterly
:::: diabolical cunning cunning cunning
:::: dick richard richard richard
:::: dictator regime regime regime
:::: die dying dying dying
:::: dies died died died
:::: difficult impossible impossible impossible
:::: difficulties difficulty difficulty difficulty
:::: difficulty difficulties difficulties difficulties
:::: dilemma confronting confronting confronting
:::: din shah shah shah
:::: dire consequences consequences consequences
:::: dirt mud mud mud
:::: dirty filthy filthy filthy
:::: disabled handicapped handicapped handicapped
:::: disagree agree agree agree
:::: disappoint disappoints disappoints disappoints
:::: disappointed pleased pleased pleased
:::: disappointing dismal dismal dismal
:::: disappointment frustration frustration frustration
:::: disappoints disappoint disappoint disappoint
:::: disaster catastrophe catastrophe catastrophe
:::: disastrous devastating devastating devastating
:::: disbelief disgust disgust disgust
:::: disdain admiration admiration admiration
:::: disgrace shame shame shame
:::: disgruntled angry angry angry
:::: disgust disbelief disbelief disbelief
:::: disgusted appalled appalled appalled
:::: disgusting revolting revolting revolting
:::: dislike disdain disdain disdain
:::: disliked hated hated hated
:::: dismal disappointing disappointing disappointing
:::: disorder traumatic traumatic traumatic
:::: disregard indifference indifference indifference
:::: distorted exaggerated exaggerated exaggerated
:::: distract distracting distracting distracting
:::: distracting distracted distracted distracted
:::: distraction distracting distracting distracting
:::: distress alarm alarm alarm
:::: disturbed profoundly profoundly profoundly
:::: disturbing frightening frightening frightening
:::: doom apocalypse apocalypse apocalypse
:::: doubt doubts doubts doubts
:::: doubts doubt doubt doubt
:::: downbeat upbeat upbeat upbeat
:::: downfall demise demise demise
:::: downhill val val val
:::: downside upside upside upside
:::: drab dreary dreary dreary
:::: drag drags drags drags

:::: dragged dragging dragging dragging
:::: dragging dragged dragged dragged
:::: drastically dramatically dramatically dramatically
:::: dread foreboding foreboding foreboding
:::: dreadful horrible horrible horrible
:::: dreadfully horribly horribly horribly
:::: dripping sweat sweat sweat
:::: drowning drowned drowned drowned
:::: drunk drunken drunken drunken
:::: drunken drunk drunk drunk
:::: dubious questionable questionable questionable
:::: dud flop flop flop
:::: dull boring boring boring
:::: dumb stupid stupid stupid
:::: dump dumped dumped dumped
:::: dumped dump dump dump
:::: dust clouds clouds clouds
:::: dusty dirt dirt dirt
:::: dying die die die
:::: eccentric quirky quirky quirky
:::: embarrass annoy annoy annoy
:::: embarrassing embarrassment embarrassment embarrassment
:::: embarrassingly shockingly shockingly shockingly
:::: embarrassment embarrassing embarrassing embarrassing
:::: enemies enemy enemy enemy
:::: enemy enemies enemies enemies
:::: error errors errors errors
:::: errors error error error
:::: estranged divorced divorced divorced
:::: exaggeration understatement understatement understatement
:::: excessive unnecessary unnecessary unnecessary
:::: excruciating painful painful painful
:::: excruciatingly painfully painfully painfully
:::: excuse excuses excuses excuses
:::: excuses excuse excuse excuse
:::: exhausted tired tired tired
:::: expensive cheaper cheaper cheaper
:::: explode exploding exploding exploding
:::: exploit exploited exploited exploited
:::: exploitation exploit exploit exploit
:::: explosive bombs bombs bombs
:::: fail fails fails fails
:::: failed failing failing failing
:::: failing failed failed failed
:::: fails fail fail fail
:::: failure failing failing failing
:::: failures failure failure failure
:::: faint glow glow glow
:::: fake phony phony phony
:::: fall falling falling falling
:::: fallen falling falling falling
:::: falling fell fell fell
:::: falls rises rises rises
:::: false misleading misleading misleading
:::: fanatic fanatics fanatics fanatics
:::: fanatics fanatic fanatic fanatic

:::: farce melodrama melodrama melodrama
:::: fascist nazi nazi nazi
:::: fat eat eat eat
:::: fatal deadly deadly deadly
:::: fault faults faults faults
:::: faults fault fault fault
:::: fears concerns concerns concerns
:::: feeble weak weak weak
:::: fell rose rose rose
:::: fever headache headache headache
:::: fiction novels novels novels
:::: fictional characters characters characters
:::: fiend hustler hustler hustler
:::: fierce intense intense intense
:::: filth filthy filthy filthy
:::: filthy dirty dirty dirty
:::: fist shaking shaking shaking
:::: flaw flaws flaws flaws
:::: flaws shortcomings shortcomings shortcomings
:::: flee fleeing fleeing fleeing
:::: fleeing flee flee flee
:::: fleeting glimpses glimpses glimpses
:::: flimsy cardboard cardboard cardboard
:::: fool fools fools fools
:::: fooled tricked tricked tricked
:::: foolish stupid stupid stupid
:::: forbidden banned banned banned
:::: foreboding ominous ominous ominous
:::: foul throws throws throws
:::: fragile shaky shaky shaky
:::: frantic frenetic frenetic frenetic
:::: fraud charges charges charges
:::: freak freaks freaks freaks
:::: freaking freaked freaked freaked
:::: freaks freak freak freak
:::: freeze frozen frozen frozen
:::: frenetic frantic frantic frantic
:::: frenzy spree spree spree
:::: fried chicken chicken chicken
:::: frightening terrifying terrifying terrifying
:::: frost winters winters winters
:::: frozen freeze freeze freeze
:::: frustrated disappointed disappointed disappointed
:::: frustrating tedious tedious tedious
:::: frustration anger anger anger
:::: fugitive arrest arrest arrest
:::: funky retro retro retro
:::: funny hilarious hilarious hilarious
:::: furious angry angry angry
:::: fury rage rage rage
:::: fuss bother bother bother
:::: gangster gangsters gangsters gangsters
:::: garbage trash trash trash
:::: gasp sigh sigh sigh
:::: ghastly horrifying horrifying horrifying
:::: ghetto nazis nazis nazis

:::: gimmick gimmicks gimmicks gimmicks
:::: gimmicks gimmick gimmick gimmick
:::: gloomy bleak bleak bleak
:::: goof goofy goofy goofy
:::: goofy silly silly silly
:::: grainy footage footage footage
:::: grating whiny whiny whiny
:::: greed arrogance arrogance arrogance
:::: greedy selfish selfish selfish
:::: grief sorrow sorrow sorrow
:::: grim bleak bleak bleak
:::: grisly gruesome gruesome gruesome
:::: gritty realism realism realism
:::: gross projected projected projected
:::: grossly exaggerated exaggerated exaggerated
:::: grotesque comical comical comical
:::: grudge rivalry rivalry rivalry
:::: gruesome grisly grisly grisly
:::: gruff demeanor demeanor demeanor
:::: grumpy dopey dopey dopey
:::: guilt innocence innocence innocence
:::: guilty convicted convicted convicted
:::: hack steal steal steal
:::: handicapped disabled disabled disabled
:::: hang hung hung hung
:::: hangs hanging hanging hanging
:::: hapless bumbling bumbling bumbling
:::: hard harder harder harder
:::: hardened criminals criminals criminals
:::: hardships endure endure endure
:::: harm damage damage damage
:::: harsh criticism criticism criticism
:::: hate hatred hatred hatred
:::: hated disliked disliked disliked
:::: hateful racist racist racist
:::: hates loves loves loves
:::: hating despise despise despise
:::: hatred hate hate hate
:::: haunt haunted haunted haunted
:::: headache pains pains pains
:::: heartless soulless soulless soulless
:::: heck yeah yeah yeah
:::: hell heaven heaven heaven
:::: helpless hopeless hopeless hopeless
:::: hideous ghastly ghastly ghastly
:::: hollow sleepy sleepy sleepy
:::: hopeless helpless helpless helpless
:::: horrendous horrific horrific horrific
:::: horrible terrible terrible terrible
:::: horrid dreadful dreadful dreadful
:::: horrific horrifying horrifying horrifying
:::: horrified appalled appalled appalled
:::: horrifying horrific horrific horrific
:::: hostage kidnapped kidnapped kidnapped
:::: hostile enemy enemy enemy
:::: hum ho ho ho

:::: hung hanging hanging hanging
:::: hurt hurting hurting hurting
:::: hurting hurt hurt hurt
:::: hurts hurting hurting hurting
:::: hustler playboy playboy playboy
:::: hype hyped hyped hyped
:::: hysterical shrill shrill shrill
:::: hysterically crying crying crying
:::: idiocy stupidity stupidity stupidity
:::: idiot idiots idiots idiots
:::: idiotic inane inane inane
:::: idiots morons morons morons
:::: ignorance indifference indifference indifference
:::: illegal immigrants immigrants immigrants
:::: illiterate ignorant ignorant ignorant
:::: illness disease disease disease
:::: illogical absurd absurd absurd
:::: imaginary fictional fictional fictional
:::: immature mature mature mature
:::: impending demise demise demise
:::: implausible preposterous preposterous preposterous
:::: impossible difficult difficult difficult
:::: improbable implausible implausible implausible
:::: inability unable unable unable
:::: inaccuracies inconsistencies inconsistencies inconsistencies
:::: inaccurate misleading misleading misleading
:::: inane idiotic idiotic idiotic
:::: inappropriate behavior behavior behavior
:::: incapable capable capable capable
:::: incoherent nonsensical nonsensical nonsensical
:::: incompetence arrogance arrogance arrogance
:::: incompetent inept inept inept
:::: incomprehensible inexplicable inexplicable inexplicable
:::: inconsistencies inaccuracies inaccuracies inaccuracies
:::: inconsistent consistent consistent consistent
:::: incorrect inaccurate inaccurate inaccurate
:::: indifference ignorance ignorance ignorance
:::: indifferent ignorant ignorant ignorant
:::: indulge fantasies fantasies fantasies
:::: inept incompetent incompetent incompetent
:::: inevitable consequence consequence consequence
:::: inexperienced talented talented talented
:::: infamous notorious notorious notorious
:::: infected virus virus virus
:::: inferior superior superior superior
:::: injury suffered suffered suffered
:::: injustice racism racism racism
:::: innuendo profanity profanity profanity
:::: insanely ridiculously ridiculously ridiculously
:::: insanity madness madness madness
:::: insecure unstable unstable unstable
:::: insignificant irrelevant irrelevant irrelevant
:::: insult insulting insulting insulting
:::: insulted offended offended offended
:::: insulting insulted insulted insulted
:::: insults shouting shouting shouting

intense fierce fierce fierce
invisible visible visible visible
ironic irony irony irony
irony ironic ironic ironic
irrational illogical illogical illogical
irrelevant meaningless meaningless meaningless
irresponsible reckless reckless reckless
irritated annoyed annoyed annoyed
irritating annoying annoying annoying
isolation loneliness loneliness loneliness
issue issues issues issues
issues issue issue issue
jaded cynical cynical cynical
jarring unsettling unsettling unsettling
jealous jealousy jealousy jealousy
jealousy envy envy envy
jerk snatch snatch snatch
jerky grainy grainy grainy
joke jokes jokes jokes
joker batman batman batman
junk trash trash trash
killed killing killing killing
killer killers killers killers
killing killed killed killed
knife knives knives knives
knock knocking knocking knocking
lack lacking lacking lacking
lacked lacking lacking lacking
lacking lacked lacked lacked
lacks lacking lacking lacking
lame duck duck duck
laughable ludicrous ludicrous ludicrous
laughably shockingly shockingly shockingly
lazy stupid stupid stupid
lengthy brief brief brief
lethal deadly deadly deadly
lie lies lies lies
lies lie lie lie
limit limits limits limits
limitations constraints constraints constraints
limits limit limit limit
limp lifeless lifeless lifeless
lone sole sole sole
lonely desolate desolate desolate
loner nerdy nerdy nerdy
longing yearning yearning yearning
loose ball ball ball
lose losing losing losing
loser losers losers losers
losers loser loser loser
loses lose lose lose
losing lost lost lost
loss losing losing losing
lost losing losing losing
loud heard heard heard
ludicrous preposterous preposterous preposterous

:::: lunatic insane insane insane
:::: lure attract attract attract
:::: lurid sensational sensational sensational
:::: lurking shadows shadows shadows
:::: lying lie lie lie
:::: macabre grotesque grotesque grotesque
:::: mad cow cow cow
:::: madness insanity insanity insanity
:::: maniac homicidal homicidal homicidal
:::: maniacal crazed crazed crazed
:::: manic psychotic psychotic psychotic
:::: manipulate manipulated manipulated manipulated
:::: manipulation manipulate manipulate manipulate
:::: manipulative conniving conniving conniving
:::: massacre killings killings killings
:::: meaningless irrelevant irrelevant irrelevant
:::: mediocre lackluster lackluster lackluster
:::: mediocrity obscurity obscurity obscurity
:::: melancholy dreamy dreamy dreamy
:::: melodramatic melodrama melodrama melodrama
:::: menace phantom phantom phantom
:::: menacing ominous ominous ominous
:::: mercilessly relentlessly relentlessly relentlessly
:::: messed screwed screwed screwed
:::: messing messed messed messed
:::: messy ugly ugly ugly
:::: mindless senseless senseless senseless
:::: mischievous playful playful playful
:::: miserable wretched wretched wretched
:::: miserably spectacularly spectacularly spectacularly
:::: misery despair despair despair
:::: misfortune luck luck luck
:::: misguided irresponsible irresponsible irresponsible
:::: misleading inaccurate inaccurate inaccurate
:::: miss missed missed missed
:::: missed misses misses misses
:::: misses missed missed missed
:::: mistake mistakes mistakes mistakes
:::: mistaken misunderstood misunderstood misunderstood
:::: mistress lover lover lover
:::: misunderstood mistaken mistaken mistaken
:::: mobster gangster gangster gangster
:::: mock staging staging staging
:::: mockery travesty travesty travesty
:::: mocking sarcastic sarcastic sarcastic
:::: monotonous tedious tedious tedious
:::: monster monsters monsters monsters
:::: monstrosity hideous hideous hideous
:::: monstrous monster monster monster
:::: moody rating rating rating
:::: morbid macabre macabre macabre
:::: moron idiot idiot idiot
:::: moronic idiotic idiotic idiotic
:::: morons idiots idiots idiots
:::: mundane banal banal banal
:::: murder murders murders murders

:::: murderer rapist rapist rapist
:::: murderous brutal brutal brutal
:::: murky shadowy shadowy shadowy
:::: mysterious strange strange strange
:::: mysteriously inexplicably inexplicably inexplicably
:::: mystery mysteries mysteries mysteries
:::: myth mythology mythology mythology
:::: naive idealistic idealistic idealistic
:::: nasty ugly ugly ugly
:::: naughty cute cute cute
:::: nauseating disgusting disgusting disgusting
:::: needless unnecessary unnecessary unnecessary
:::: needlessly unnecessarily unnecessarily unnecessarily
:::: negative positive positive positive
:::: neglect neglected neglected neglected
:::: neglected forgotten forgotten forgotten
:::: nemesis villain villain villain
:::: nervous anxious anxious anxious
:::: neurotic obsessive obsessive obsessive
:::: nightmare nightmarish nightmarish nightmarish
:::: nightmarish nightmare nightmare nightmare
:::: noise noises noises noises
:::: noises loud loud loud
:::: nonexistent existent existent existent
:::: nonsense ridiculous ridiculous ridiculous
:::: notorious infamous infamous infamous
:::: object objects objects objects
:::: oblivious unaware unaware unaware
:::: obnoxious annoying annoying annoying
:::: obscene vulgar vulgar vulgar
:::: obscure origins origins origins
:::: obscurity oblivion oblivion oblivion
:::: obsessive obsession obsession obsession
:::: odd strange strange strange
:::: oddly strangely strangely strangely
:::: offend offended offended offended
:::: offensive offense offense offense
:::: ominous unsettling unsettling unsettling
:::: opponent opponents opponents opponents
:::: opposition party party party
:::: oppressive exploitative exploitative exploitative
:::: ordeal harrowing harrowing harrowing
:::: orphan orphanage orphanage orphanage
:::: outlaw biker biker biker
:::: outrage anger anger anger
:::: outrageous ridiculous ridiculous ridiculous
:::: outrageously ridiculously ridiculously ridiculously
:::: overbearing obnoxious obnoxious obnoxious
:::: overblown exaggerated exaggerated exaggerated
:::: overdone overblown overblown overblown
:::: overrated underrated underrated underrated
:::: overwhelmed exhausted exhausted exhausted
:::: overwhelming majority majority majority
:::: pain suffering suffering suffering
:::: painful wrenching wrenching wrenching
:::: painfully excruciatingly excruciatingly excruciatingly

:::: pains complaining complaining complaining
:::: pale dark dark dark
:::: pan pans pans pans
:::: panic fear fear fear
:::: paranoia paranoid paranoid paranoid
:::: paranoid paranoia paranoia paranoia
:::: parody parodies parodies parodies
:::: passive active active active
:::: pathetic pitiful pitiful pitiful
:::: peculiar strange strange strange
:::: penalty kick kick kick
:::: peril danger danger danger
:::: perverse illogical illogical illogical
:::: petty crooks crooks crooks
:::: phony bogus bogus bogus
:::: pig pigs pigs pigs
:::: pigs pig pig pig
:::: pitiful pathetic pathetic pathetic
:::: pity shame shame shame
:::: plague disease disease disease
:::: plight hardships hardships hardships
:::: plot plots plots plots
:::: poison lethal lethal lethal
:::: pompous pretentious pretentious pretentious
:::: poorly badly badly badly
:::: poverty poor poor poor
:::: prejudice prejudices prejudices prejudices
:::: prejudices stereotypes stereotypes stereotypes
:::: preposterous ludicrous ludicrous ludicrous
:::: pretend pretending pretending pretending
:::: pretentious pompous pompous pompous
:::: primitive evolved evolved evolved
:::: prison jail jail jail
:::: prisoner prisoners prisoners prisoners
:::: problem problems problems problems
:::: problems problem problem problem
:::: profanity nudity nudity nudity
:::: propaganda ideology ideology ideology
:::: protest opposition opposition opposition
:::: provocative irresponsible irresponsible irresponsible
:::: punch punches punches punches
:::: punk hardcore hardcore hardcore
:::: puppet puppets puppets puppets
:::: puppets puppet puppet puppet
:::: puzzled annoyed annoyed annoyed
:::: puzzling puzzled puzzled puzzled
:::: questionable dubious dubious dubious
:::: racism prejudice prejudice prejudice
:::: racist sexist sexist sexist
:::: radical movement movement movement
:::: rage anger anger anger
:::: raging battling battling battling
:::: rampage spree spree spree
:::: rampant corruption corruption corruption
:::: randomly random random random
:::: rant rave rave rave

:::: rape murder murder murder
:::: raped murdered murdered murdered
:::: raving loony loony loony
:::: rebellious rebellion rebellion rebellion
:::: reckless irresponsible irresponsible irresponsible
:::: redundant superfluous superfluous superfluous
:::: refuse refused refused refused
:::: refused asked asked asked
:::: refuses agrees agrees agrees
:::: regret regrets regrets regrets
:::: regrets regret regret regret
:::: reject rejected rejected rejected
:::: rejected reject reject reject
:::: relentless intense intense intense
:::: relentlessly mercilessly mercilessly mercilessly
:::: reluctant unwilling unwilling unwilling
:::: reluctantly willingly willingly willingly
:::: repetitive monotonous monotonous monotonous
:::: resistance forces forces forces
:::: restless weary weary weary
:::: retarded mentally mentally mentally
:::: revenge vengeance vengeance vengeance
:::: revolt rebellion rebellion rebellion
:::: revolting disgusting disgusting disgusting
:::: ridiculous absurd absurd absurd
:::: rigid strict strict strict
:::: rip ripping ripping ripping
:::: ripoff stinker stinker stinker
:::: ripped ripping ripping ripping
:::: risk risks risks risks
:::: risks risk risk risk
:::: rival rivals rivals rivals
:::: rivalry bitter bitter bitter
:::: rocky mountain mountain mountain
:::: rogue terrorists terrorists terrorists
:::: rotten tomatoes tomatoes tomatoes
:::: rough tough tough tough
:::: rubbish garbage garbage garbage
:::: rude obnoxious obnoxious obnoxious
:::: ruin ruined ruined ruined
:::: ruined ruin ruin ruin
:::: ruining spoiling spoiling spoiling
:::: ruins ruined ruined ruined
:::: rumors reports reports reports
:::: runaway teenage teenage teenage
:::: rusty nails nails nails
:::: ruthless brutal brutal brutal
:::: sack threw threw threw
:::: sacrificed sacrifice sacrifice sacrifice
:::: sad sorry sorry sorry
:::: sadly unfortunately unfortunately unfortunately
:::: sadness sorrow sorrow sorrow
:::: sarcasm irony irony irony
:::: satirical satire satire satire
:::: savage brutal brutal brutal
:::: scandal revelations revelations revelations

:::: scare scares scares scares
:::: scared frightened frightened frightened
:::: scarier funnier funnier funnier
:::: scariest strangest strangest strangest
:::: scary frightening frightening frightening
:::: scratch scratching scratching scratching
:::: scream shout shout shout
:::: screwed messed messed messed
:::: seedy sleazy sleazy sleazy
:::: selfish greedy greedy greedy
:::: senseless cowardly cowardly cowardly
:::: seriousness sincerity sincerity sincerity
:::: severe suffering suffering suffering
:::: shady dodgy dodgy dodgy
:::: shake shaking shaking shaking
:::: shaky fragile fragile fragile
:::: shallow deep deep deep
:::: shame disgrace disgrace disgrace
:::: shameful despicable despicable despicable
:::: shameless blatant blatant blatant
:::: shamelessly blatantly blatantly blatantly
:::: shark whale whale whale
:::: shock shocks shocks shocks
:::: shocked stunned stunned stunned
:::: shocking startling startling startling
:::: shockingly embarrassingly embarrassingly embarrassingly
:::: shoddy sloppy sloppy sloppy
:::: shortcomings flaws flaws flaws
:::: showdown confrontation confrontation confrontation
:::: shrill hysterical hysterical hysterical
:::: sick ill ill ill
:::: sickening horrifying horrifying horrifying
:::: sickness illness illness illness
:::: siege battle battle battle
:::: silly stupid stupid stupid
:::: sin sins sins sins
:::: sinister evil evil evil
:::: sink sinks sinks sinks
:::: sinking sunk sunk sunk
:::: skeptical convinced convinced convinced
:::: skimpy outfits outfits outfits
:::: skinny pants pants pants
:::: slap slapped slapped slapped
:::: slaughter slaughtered slaughtered slaughtered
:::: slaughtered slaughter slaughter slaughter
:::: slave slaves slaves slaves
:::: slaves slave slave slave
:::: sloppy shoddy shoddy shoddy
:::: slower faster faster faster
:::: slowly gradually gradually gradually
:::: slug slugs slugs slugs
:::: slut bitch bitch bitch
:::: sly wit wit wit
:::: smash hit hit hit
:::: smell taste taste taste
:::: smoke smoking smoking smoking

:::: smug cocky cocky cocky
:::: sneak preview preview preview
:::: sober thoughtful thoughtful thoughtful
:::: sore shoulder shoulder shoulder
:::: sorrow sadness sadness sadness
:::: sorry ashamed ashamed ashamed
:::: sour taste taste taste
:::: spade kate kate kate
:::: spite despite despite despite
:::: splatter slasher slasher slasher
:::: split divided divided divided
:::: splitting split split split
:::: spoil spoiling spoiling spoiling
:::: spoiled spoiling spoiling spoiling
:::: spooky creepy creepy creepy
:::: stab wounds wounds wounds
:::: stale tiresome tiresome tiresome
:::: stark reminder reminder reminder
:::: startling surprising surprising surprising
:::: static dynamic dynamic dynamic
:::: steal stole stole stole
:::: stealing stole stole stole
:::: steals throws throws throws
:::: stereotype stereotypes stereotypes stereotypes
:::: stereotypical stereotype stereotype stereotype
:::: stern howard howard howard
:::: stiff face face face
:::: stink smell smell smell
:::: stinks sucks sucks sucks
:::: stole steal steal steal
:::: stolen stole stole stole
:::: stooges slapstick slapstick slapstick
:::: strain virus virus virus
:::: strained sore sore sore
:::: strange weird weird weird
:::: strangely oddly oddly oddly
:::: stranger strangers strangers strangers
:::: strangest saddest saddest saddest
:::: stress anxiety anxiety anxiety
:::: strict rules rules rules
:::: strictly strict strict strict
:::: strike strikes strikes strikes
:::: struck hit hit hit
:::: struggle struggles struggles struggles
:::: struggles struggle struggle struggle
:::: struggling struggles struggles struggles
:::: stubborn overcome overcome overcome
:::: stuck sticking sticking sticking
:::: stumble stumbles stumbles stumbles
:::: stumbled stumble stumble stumble
:::: stumbles stumble stumble stumble
:::: stunt stunts stunts stunts
:::: stupid silly silly silly
:::: stupidest dumbest dumbest dumbest
:::: stupidity arrogance arrogance arrogance
:::: subdued restrained restrained restrained

:::: subjected endure endure endure
:::: sucked suck suck suck
:::: sucker punch punch punch
:::: sucks suck suck suck
:::: sue suit suit suit
:::: suffer suffering suffering suffering
:::: suffered suffering suffering suffering
:::: suffering suffered suffered suffered
:::: suffers suffer suffer suffer
:::: suicide attacks attacks attacks
:::: sunk sinking sinking sinking
:::: superficial resemblance resemblance resemblance
:::: superfluous useless useless useless
:::: suspect suspects suspects suspects
:::: suspicion arrested arrested arrested
:::: suspicious suspect suspect suspect
:::: syndrome disorder disorder disorder
:::: tacky cheesy cheesy cheesy
:::: tank rocket rocket rocket
:::: taut suspenseful suspenseful suspenseful
:::: tedious tiresome tiresome tiresome
:::: temper fiery fiery fiery
:::: temptation resist resist resist
:::: tense uneasy uneasy uneasy
:::: tension tensions tensions tensions
:::: tepid lackluster lackluster lackluster
:::: terrible horrible horrible horrible
:::: terror terrorist terrorist terrorist
:::: terrorism terror terror terror
:::: threat threats threats threats
:::: threatening threatened threatened threatened
:::: threats threat threat threat
:::: thug gangster gangster gangster
:::: tired weary weary weary
:::: tiresome tedious tedious tedious
:::: torment anguish anguish anguish
:::: tormented torment torment torment
:::: torture tortured tortured tortured
:::: tortured raped raped raped
:::: torturing murdering murdering murdering
:::: tragedy tragic tragic tragic
:::: tragic tragedy tragedy tragedy
:::: tramp chaplin chaplin chaplin
:::: trap traps traps traps
:::: trapped rescued rescued rescued
:::: trash garbage garbage garbage
:::: trauma traumatic traumatic traumatic
:::: traumatic trauma trauma trauma
:::: traumatized frightened frightened frightened
:::: travesty mockery mockery mockery
:::: trick tricks tricks tricks
:::: tricked fooled fooled fooled
:::: trivial mundane mundane mundane
:::: troubled struggling struggling struggling
:::: troubles problems problems problems
:::: turmoil crisis crisis crisis

:::: twist twists twists twists
:::: twisted broken broken broken
:::: twists twist twist twist
:::: ugh hmmm hmmm hmmm
:::: ugly nasty nasty nasty
:::: ultimatum surrender surrender surrender
:::: unable able able able
:::: unappealing unattractive unattractive unattractive
:::: unattractive unappealing unappealing unappealing
:::: unbearable excruciating excruciating excruciating
:::: unbelievable incredible incredible incredible
:::: unbelievably incredibly incredibly incredibly
:::: uncertain unclear unclear unclear
:::: unclear uncertain uncertain uncertain
:::: uncomfortable awkward awkward awkward
:::: unconvincing unimpressive unimpressive unimpressive
:::: underdog favorites favorites favorites
:::: uneasy tense tense tense
:::: uneven slow slow slow
:::: unexpected surprising surprising surprising
:::: unexpectedly surprisingly surprisingly surprisingly
:::: unexplained inexplicable inexplicable inexplicable
:::: unfairly unfair unfair unfair
:::: unfamiliar familiar familiar familiar
:::: unfinished monumental monumental monumental
:::: unfortunate tragic tragic tragic
:::: unlikely likely likely likely
:::: unlucky fortunate fortunate fortunate
:::: unnatural inexplicable inexplicable inexplicable
:::: unnecessary needless needless needless
:::: unnerving unsettling unsettling unsettling
:::: unpredictable unstable unstable unstable
:::: unravel unfold unfold unfold
:::: unrealistic illogical illogical illogical
:::: unsettling unnerving unnerving unnerving
:::: unstable dangerously dangerously dangerously
:::: unsuccessful successful successful successful
:::: unsure uncertain uncertain uncertain
:::: unsuspecting lure lure lure
:::: untrue inaccurate inaccurate inaccurate
:::: unusual rare rare rare
:::: unusually exceptionally exceptionally exceptionally
:::: unwatchable watchable watchable watchable
:::: unwilling reluctant reluctant reluctant
:::: upset stunned stunned stunned
:::: useless meaningless meaningless meaningless
:::: utterly totally totally totally
:::: vague ambiguous ambiguous ambiguous
:::: vain desperately desperately desperately
:::: vanity esquire esquire esquire
:::: vengeance revenge revenge revenge
:::: vengeful murderous murderous murderous
:::: vice president president president
:::: vicious brutal brutal brutal
:::: vile despicable despicable despicable
:::: villainous villain villain villain


```

:::: villains villain villain villain
:::: violent violence violence violence
:::: violently brutally brutally brutally
:::: virus infected infected infected
:::: vomit suck suck suck
:::: vulgar obscene obscene obscene
:::: vulnerable vulnerability vulnerability vulnerability
:::: warned warn warn warn
:::: warning warn warn warn
:::: waste wastes wastes wastes
:::: wasting wasted wasted wasted
:::: weak weaker weaker weaker
:::: weaker stronger stronger stronger
:::: weakness weak weak weak
:::: weaknesses shortcomings shortcomings shortcomings
:::: weary tired tired tired
:::: weird strange strange strange
:::: whining whiny whiny whiny
:::: whore slut slut slut
:::: wicked witch witch witch
:::: wild animals animals animals
:::: wildly hugely hugely hugely
:::: woefully shockingly shockingly shockingly
:::: worn wore wore wore
:::: worry worried worried worried
:::: worrying worried worried worried
:::: worst worse worse worse
:::: worthless useless useless useless
:::: wound wounds wounds wounds
:::: wounds wound wound wound
:::: wrath fury fury fury
:::: wreck salvage salvage salvage
:::: wretched miserable miserable miserable
:::: wrongly rightly rightly rightly
:::: yawn sigh sigh sigh
:::: zombie zombies zombies zombies
1716

```

12. t-SNE Visualization

Project positive and negative lexicon words to 2D for GloVe, dense, and LSTM embeddings.

```

In [ ]: emb_glove = []
emb_dense = []
emb_LSTM = []
for word in L_POS + L_NEG:
    emb = embedding_matrix[word_index[word]]
    emb_glove.append(emb)
    emb = embedding_matrix1[word_index[word]]
    emb_dense.append(emb)
    emb = embedding_matrix2[word_index[word]]
    emb_LSTM.append(emb)

```

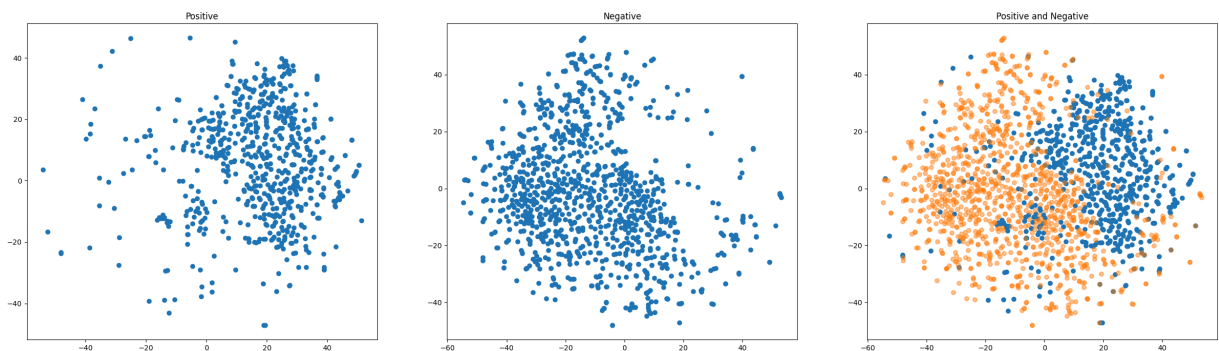
```
In [ ]: emb_glove = np.array(emb_glove)
emb_dense = np.array(emb_dense)
emb_LSTM = np.array(emb_LSTM)
```

```
In [ ]: emb_glove.shape, emb_dense.shape, emb_LSTM.shape
```

```
Out[ ]: ((1870, 300), (1870, 300), (1870, 300))
```

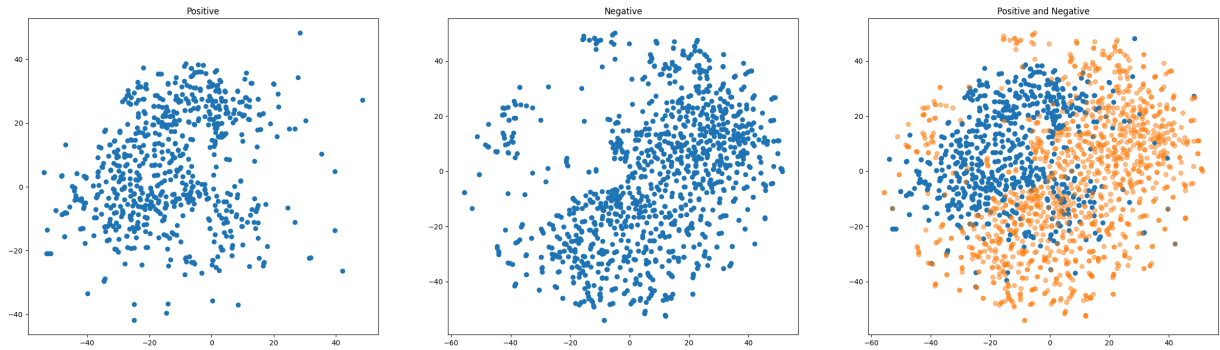
```
In [ ]: import numpy as np
from sklearn.manifold import TSNE
X_LSTM = TSNE(n_components=2).fit_transform(emb_LSTM)
```

```
In [ ]: import matplotlib.pyplot as plt
plt.figure(figsize=(30,8))
plt.subplot(131)
plt.scatter(X_LSTM[:len(L_POS),0], X_LSTM[:len(L_POS),1])
plt.title("Positive")
plt.subplot(132)
plt.scatter(X_LSTM[len(L_POS):,0], X_LSTM[len(L_POS):,1])
plt.title("Negative")
plt.subplot(133)
plt.scatter(X_LSTM[:len(L_POS),0], X_LSTM[:len(L_POS),1])
plt.scatter(X_LSTM[len(L_POS):,0], X_LSTM[len(L_POS):,1], alpha=0.5)
plt.title("Positive and Negative")
plt.show()
```



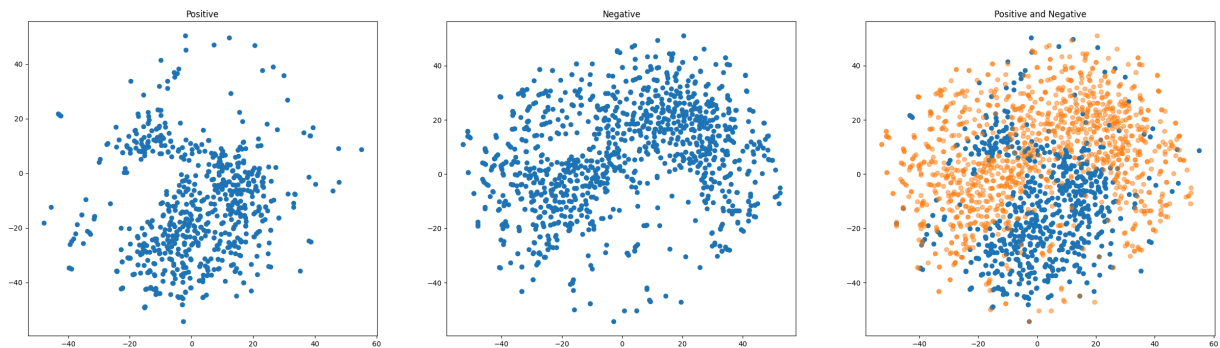
```
In [ ]: import numpy as np
from sklearn.manifold import TSNE
X_dense = TSNE(n_components=2).fit_transform(emb_dense)
```

```
In [ ]: import matplotlib.pyplot as plt
plt.figure(figsize=(30,8))
plt.subplot(131)
plt.scatter(X_dense[:len(L_POS),0], X_dense[:len(L_POS),1])
plt.title("Positive")
plt.subplot(132)
plt.scatter(X_dense[len(L_POS):,0], X_dense[len(L_POS):,1])
plt.title("Negative")
plt.subplot(133)
plt.scatter(X_dense[:len(L_POS),0], X_dense[:len(L_POS),1])
plt.scatter(X_dense[len(L_POS):,0], X_dense[len(L_POS):,1], alpha=0.5)
plt.title("Positive and Negative")
plt.show()
```



```
In [ ]: import numpy as np
        from sklearn.manifold import TSNE
        X_glove = TSNE(n_components=2).fit_transform(emb_glove)
```

```
In [ ]: import matplotlib.pyplot as plt
        plt.figure(figsize=(30,8))
        plt.subplot(131)
        plt.scatter(X_glove[:len(L_POS),0], X_glove[:len(L_POS),1])
        plt.title("Positive")
        plt.subplot(132)
        plt.scatter(X_glove[len(L_POS):,0], X_glove[len(L_POS):,1])
        plt.title("Negative")
        plt.subplot(133)
        plt.scatter(X_glove[:len(L_POS),0], X_glove[:len(L_POS),1])
        plt.scatter(X_glove[len(L_POS):,0], X_glove[len(L_POS):,1], alpha=0.5)
        plt.title("Positive and Negative")
        plt.show()
```



13. Class Separation Metric

Ratio of average cross-class distance to average within-class distance — lower is better separation.

```
In [ ]: def class_dist(L_POS, L_NEG, embedding_matrix):
        def get_vec(word):
            if word in word_index and word_index[word] < max_words:
                return embedding_matrix[word_index[word]]
            return None
        pos_vecs = [get_vec(w) for w in L_POS if get_vec(w) is not None]
        neg_vecs = [get_vec(w) for w in L_NEG if get_vec(w) is not None]
        pos_vecs = np.array(pos_vecs)
```

```

neg_vecs = np.array(neg_vecs)
d12 = np.mean([
    np.linalg.norm(p - n)
    for p in pos_vecs for n in neg_vecs
])
all_vecs = np.vstack((pos_vecs, neg_vecs))
d = np.mean([
    np.linalg.norm(a - b)
    for i, a in enumerate(all_vecs)
    for j, b in enumerate(all_vecs)
    if i < j
])
return d12 / d
def class_dist_glove():
    return class_dist(L_POS, L_NEG, embedding_matrix)
def class_dist_dense():
    return class_dist(L_POS, L_NEG, embedding_matrix1)
def class_dist_LSTM():
    return class_dist(L_POS, L_NEG, embedding_matrix2)
d_glove = class_dist_glove()
d_dense = class_dist_dense()
d_LSTM = class_dist_LSTM()

```

```

In [ ]: print(d_glove, d_dense, d_LSTM)
print("Relative improvements:")
print("Dense:", (d_dense-d_glove)/d_glove*100)
print("LSTM:", (d_LSTM-d_glove)/d_glove*100)

```

```

1.009301743943346 1.010387 1.0096298
Relative improvements:
Dense: 0.10751986512504677
LSTM: 0.03250778846113197

```

Confirm the best embedding (LSTM-tuned) improves validation accuracy when used in the LSTM classifier.

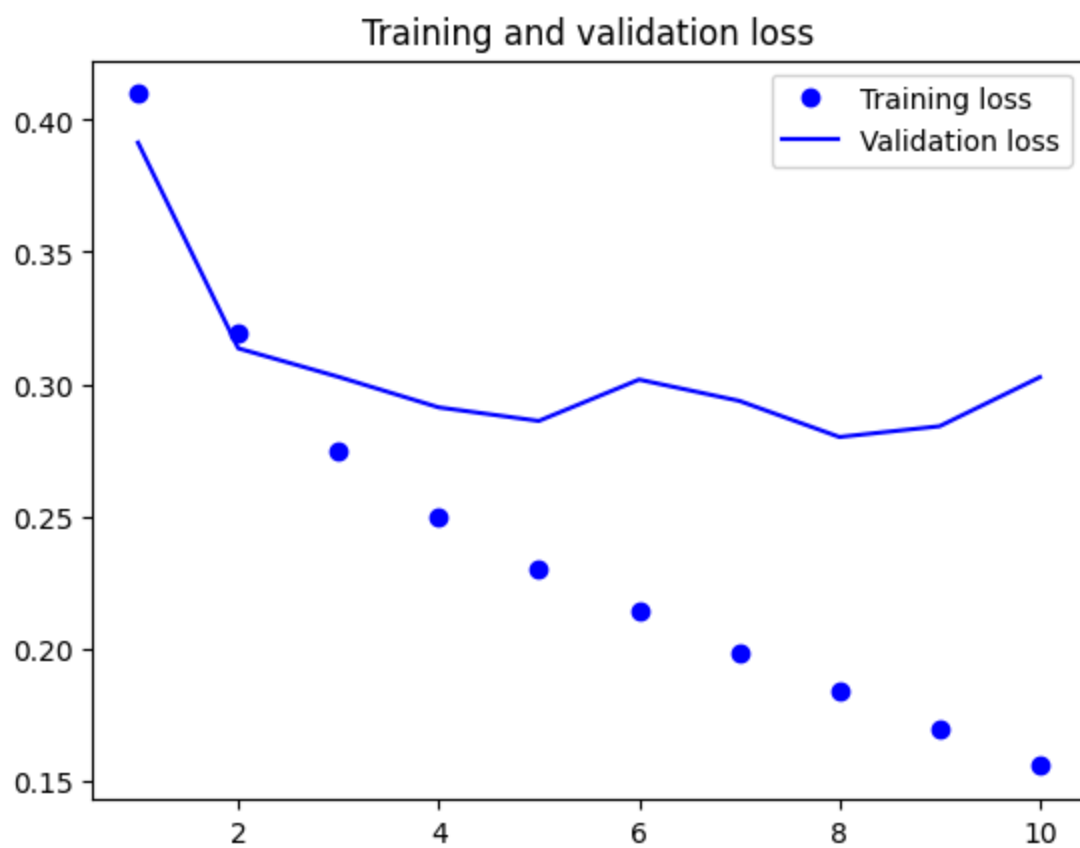
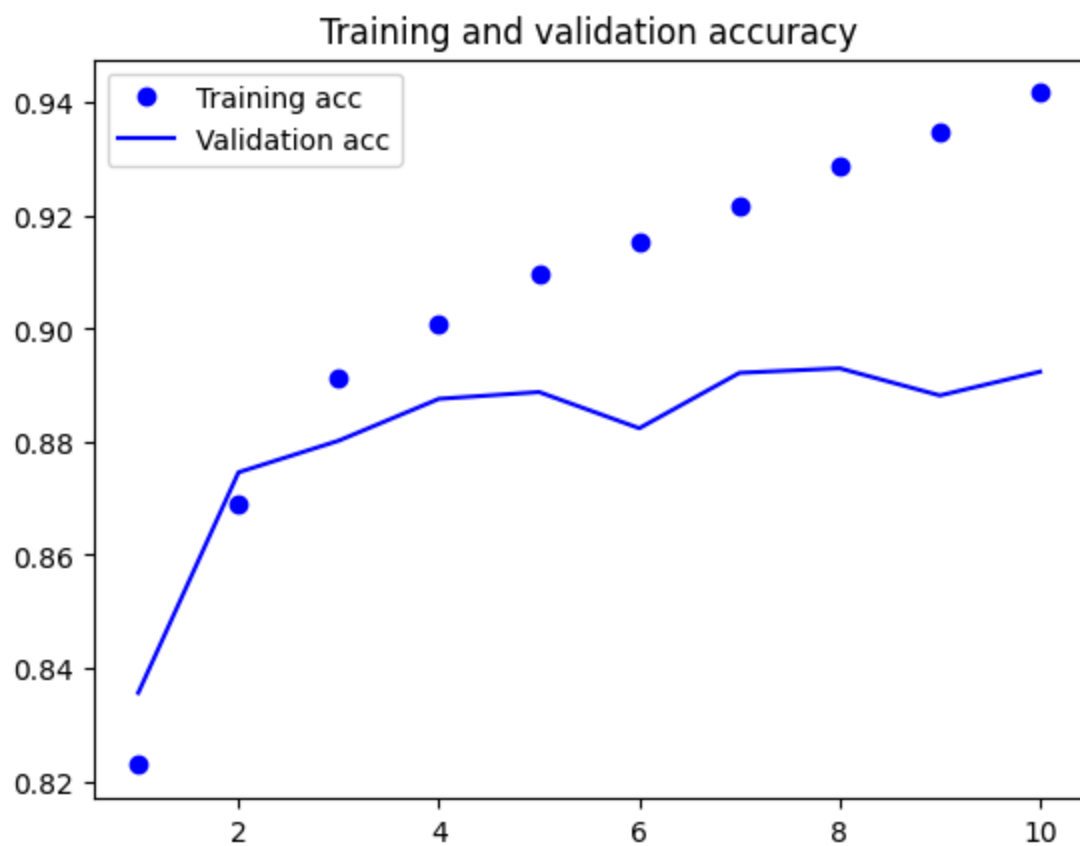
```

In [ ]: from keras.models import Sequential
from keras.layers import Embedding, LSTM, Dense
model = Sequential()
embedding_layer = Embedding(input_dim=max_words,
                             output_dim=embedding_dim,
                             weights=[embedding_matrix1],
                             trainable=False)
model.add(embedding_layer)
model.add(LSTM(32))
model.add(Dense(1, activation='sigmoid'))
model.compile(optimizer='rmsprop',
              loss='binary_crossentropy',
              metrics=['acc'])
history = model.fit(x_train, y_train,
                    epochs=10,
                    batch_size=32,
                    validation_data=(x_val, y_val))

```

Epoch 1/10
625/625  **13s** 18ms/step - acc: 0.7842 - loss: 0.4646 - val_acc: 0.8356 - val_loss: 0.3914
Epoch 2/10
625/625  **12s** 19ms/step - acc: 0.8610 - loss: 0.3359 - val_acc: 0.8746 - val_loss: 0.3135
Epoch 3/10
625/625  **12s** 19ms/step - acc: 0.8879 - loss: 0.2839 - val_acc: 0.8802 - val_loss: 0.3027
Epoch 4/10
625/625  **12s** 19ms/step - acc: 0.9020 - loss: 0.2488 - val_acc: 0.8876 - val_loss: 0.2912
Epoch 5/10
625/625  **11s** 17ms/step - acc: 0.9104 - loss: 0.2318 - val_acc: 0.8888 - val_loss: 0.2861
Epoch 6/10
625/625  **28s** 29ms/step - acc: 0.9179 - loss: 0.2100 - val_acc: 0.8824 - val_loss: 0.3017
Epoch 7/10
625/625  **14s** 19ms/step - acc: 0.9221 - loss: 0.2011 - val_acc: 0.8922 - val_loss: 0.2937
Epoch 8/10
625/625  **21s** 19ms/step - acc: 0.9303 - loss: 0.1775 - val_acc: 0.8930 - val_loss: 0.2800
Epoch 9/10
625/625  **12s** 19ms/step - acc: 0.9351 - loss: 0.1648 - val_acc: 0.8882 - val_loss: 0.2841
Epoch 10/10
625/625  **11s** 18ms/step - acc: 0.9449 - loss: 0.1537 - val_acc: 0.8924 - val_loss: 0.3026

In []: `plot_history(history)`



14. Y-Network Architecture

Dual-input model: two parallel BiLSTM → Conv1D branches on the same review, concatenated before dense layers.

Both branches share architecture but have independent embedding layers for staged fine-tuning.

```
In [ ]: from keras import backend as K
        K.clear_session()
```

```
In [ ]: import keras
        from keras import layers
        from keras import models
        input_shape = (maxlen, )
        left_inputs = layers.Input(shape=input_shape)
        x = Embedding(max_words, embedding_dim, weights=[embedding_matrix1], trainable=True)(left_inputs)
        x = layers.Bidirectional(layers.LSTM(128, return_sequences = True))(x)
        x = layers.Conv1D(64, 7, activation='relu')(x)
        x = layers.GlobalMaxPool1D()(x)
        right_inputs = layers.Input(shape=input_shape)
        y = layers.Embedding(max_words, embedding_dim, weights=[embedding_matrix1], trainable=True)(right_inputs)
        y = layers.Bidirectional(layers.LSTM(128, return_sequences = True))(y)
        y = layers.Conv1D(64, 7, activation='relu')(y)
        y = layers.GlobalMaxPool1D()(y)
        x = layers.concatenate([x, y])
        x = layers.Flatten()(x)
        x = layers.Dense(512, activation='relu')(x)
        x = layers.Dropout(0.5)(x)
        outputs = layers.Dense(1, activation='sigmoid')(x)
        model = models.Model([left_inputs, right_inputs], outputs=outputs)
```

```
/usr/local/lib/python3.12/dist-packages/keras/src/layers/core/embedding.py:97: UserWarning: Argument `input_length` is deprecated. Just remove it.
  warnings.warn(
```

```
In [ ]: model.summary()
```

Model: "functional"

Layer (type)	Output Shape	Param #	Connected to
input_layer (InputLayer)	(None, 500)	0	–
input_layer_1 (InputLayer)	(None, 500)	0	–
embedding (Embedding)	(None, 500, 300)	3,000,000	input_layer[0]
embedding_1 (Embedding)	(None, 500, 300)	3,000,000	input_layer_1[0]
bidirectional (Bidirectional)	(None, 500, 256)	439,296	embedding[0][0]
bidirectional_1 (Bidirectional)	(None, 500, 256)	439,296	embedding_1[0][0]
conv1d (Conv1D)	(None, 494, 64)	114,752	bidirectional[0][0]
conv1d_1 (Conv1D)	(None, 494, 64)	114,752	bidirectional_1[0][0]
global_max_pooling... (GlobalMaxPooling1D)	(None, 64)	0	conv1d[0][0]
global_max_pooling... (GlobalMaxPooling1D)	(None, 64)	0	conv1d_1[0][0]
concatenate (Concatenate)	(None, 128)	0	global_max_po global_max_po
flatten (Flatten)	(None, 128)	0	concatenate[0]
dense (Dense)	(None, 512)	66,048	flatten[0][0]
dropout (Dropout)	(None, 512)	0	dense[0][0]
dense_1 (Dense)	(None, 1)	513	dropout[0][0]

Total params: 7,174,657 (27.37 MB)

Trainable params: 1,174,657 (4.48 MB)

Non-trainable params: 6,000,000 (22.89 MB)

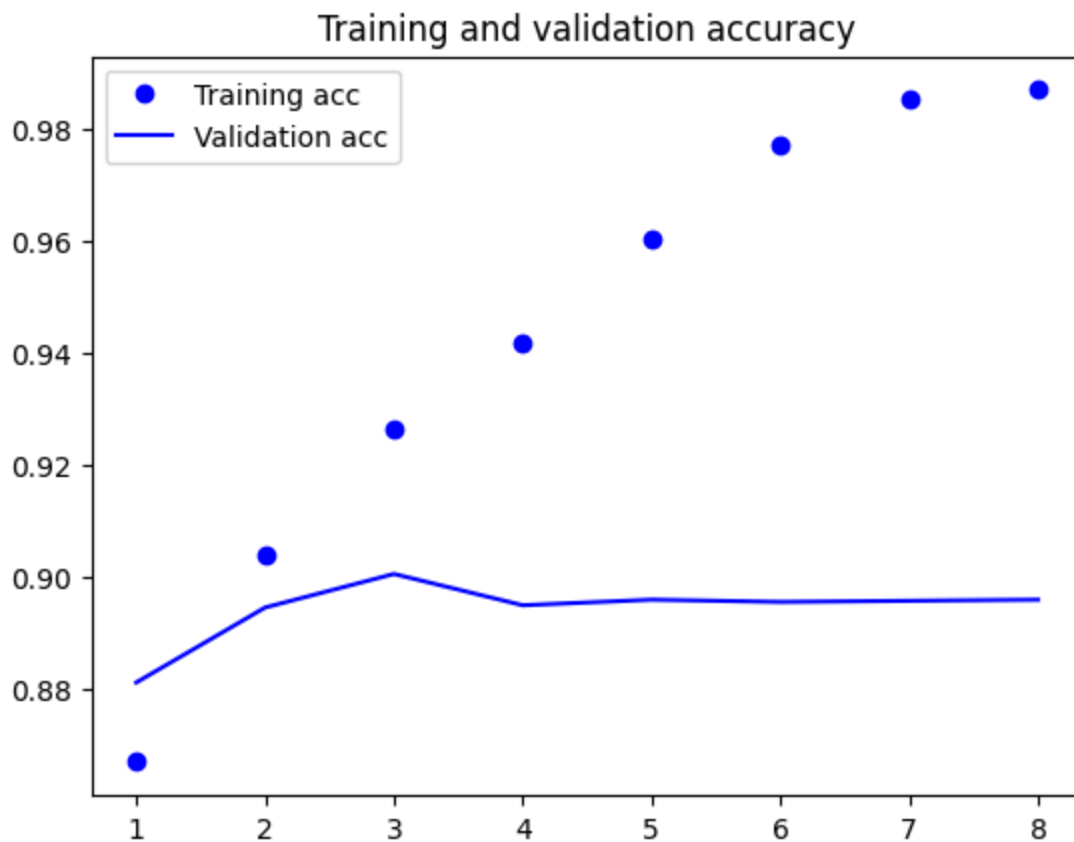
```
In [ ]: callbacks_list = [
        keras.callbacks.EarlyStopping(monitor='val_loss', patience=5,),
        keras.callbacks.ModelCheckpoint(filepath='LSTM_Conv1D.keras', monitor='v
model.compile(optimizer='adam',
               loss='binary_crossentropy',
               metrics=['acc'])
history = model.fit([x_train, x_train], y_train, shuffle=True,
                   epochs=200,
                   batch_size=32,
                   callbacks=callbacks_list,
                   validation_data=([x_val, x_val], y_val))
```

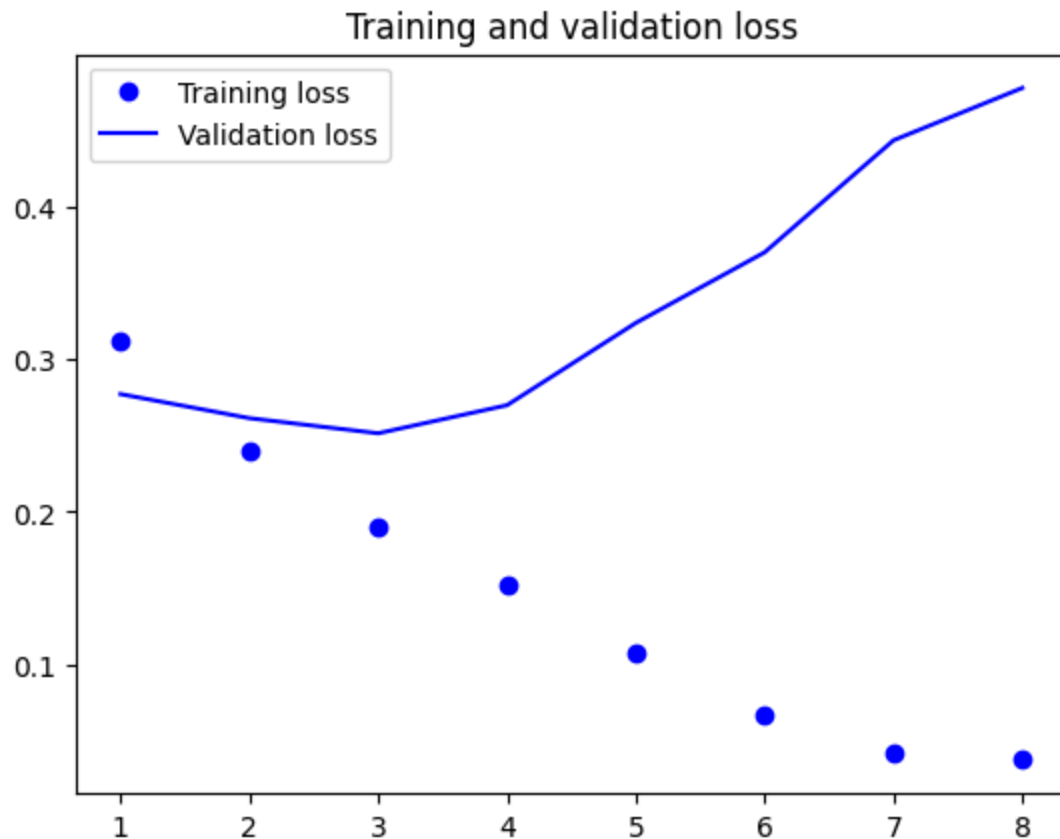

Epoch 1/200
625/625 ████████████████████ 67s 96ms/step - acc: 0.8198 - loss: 0.3794 - val_acc: 0.8810 - val_loss: 0.2771
Epoch 2/200
625/625 ████████████████████ 59s 94ms/step - acc: 0.8995 - loss: 0.2428 - val_acc: 0.8944 - val_loss: 0.2613
Epoch 3/200
625/625 ████████████████████ 58s 93ms/step - acc: 0.9273 - loss: 0.1877 - val_acc: 0.9004 - val_loss: 0.2514
Epoch 4/200
625/625 ████████████████████ 58s 93ms/step - acc: 0.9426 - loss: 0.1441 - val_acc: 0.8948 - val_loss: 0.2698
Epoch 5/200
625/625 ████████████████████ 58s 93ms/step - acc: 0.9647 - loss: 0.1000 - val_acc: 0.8958 - val_loss: 0.3236
Epoch 6/200
625/625 ████████████████████ 58s 93ms/step - acc: 0.9798 - loss: 0.0603 - val_acc: 0.8954 - val_loss: 0.3698
Epoch 7/200
625/625 ████████████████████ 58s 93ms/step - acc: 0.9883 - loss: 0.0369 - val_acc: 0.8956 - val_loss: 0.4430
Epoch 8/200
625/625 ████████████████████ 58s 93ms/step - acc: 0.9892 - loss: 0.0318 - val_acc: 0.8958 - val_loss: 0.4772

```
In [ ]: print(model.layers[3].name)
```

embedding_1

```
In [ ]: plot_history(history)
```





Phase 2: Fine-tune left-branch embedding

```
In [ ]: model = keras.models.load_model('LSTM_Conv1D.keras')
for i in range(len(model.layers)):
    model.layers[i].trainable = False
model.layers[2].trainable = True
model.compile(optimizer='adam',
              loss='binary_crossentropy',
              metrics=['acc'])
model.summary()
```

Model: "functional"

Layer (type)	Output Shape	Param #	Connected to
input_layer (InputLayer)	(None, 500)	0	–
input_layer_1 (InputLayer)	(None, 500)	0	–
embedding (Embedding)	(None, 500, 300)	3,000,000	input_layer[0]
embedding_1 (Embedding)	(None, 500, 300)	3,000,000	input_layer_1[0]
bidirectional (Bidirectional)	(None, 500, 256)	439,296	embedding[0][0]
bidirectional_1 (Bidirectional)	(None, 500, 256)	439,296	embedding_1[0][0]
conv1d (Conv1D)	(None, 494, 64)	114,752	bidirectional[0][0]
conv1d_1 (Conv1D)	(None, 494, 64)	114,752	bidirectional_1[0][0]
global_max_pooling1d (GlobalMaxPooling1D)	(None, 64)	0	conv1d[0][0]
global_max_pooling1d_1 (GlobalMaxPooling1D)	(None, 64)	0	conv1d_1[0][0]
concatenate (Concatenate)	(None, 128)	0	global_max_pooling1d[0] global_max_pooling1d_1[0]
flatten (Flatten)	(None, 128)	0	concatenate[0]
dense (Dense)	(None, 512)	66,048	flatten[0][0]
dropout (Dropout)	(None, 512)	0	dense[0][0]
dense_1 (Dense)	(None, 1)	513	dropout[0][0]

Total params: 7,174,657 (27.37 MB)

Trainable params: 3,000,000 (11.44 MB)

Non-trainable params: 4,174,657 (15.93 MB)

```
In [ ]: callbacks_list = [
        keras.callbacks.EarlyStopping(monitor='val_loss', patience=5,),
        keras.callbacks.ModelCheckpoint(filepath='LSTM_Conv1D_retrain.keras', mc
history = model.fit([x_train, x_train], y_train, shuffle=True,
                    epochs=200,
                    batch_size=32,
                    callbacks=callbacks_list,
                    validation_data=(x_val, y_val))
```

Epoch 1/200

625/625 ————— **47s** 72ms/step - acc: 0.9541 - loss: 0.1321 - val_acc: 0.9062 - val_loss: 0.2336

Epoch 2/200

625/625 ————— **44s** 71ms/step - acc: 0.9789 - loss: 0.0835 - val_acc: 0.9038 - val_loss: 0.2394

Epoch 3/200

625/625 ————— **44s** 71ms/step - acc: 0.9897 - loss: 0.0550 - val_acc: 0.9054 - val_loss: 0.2451

Epoch 4/200

625/625 ————— **44s** 71ms/step - acc: 0.9947 - loss: 0.0388 - val_acc: 0.9044 - val_loss: 0.2536

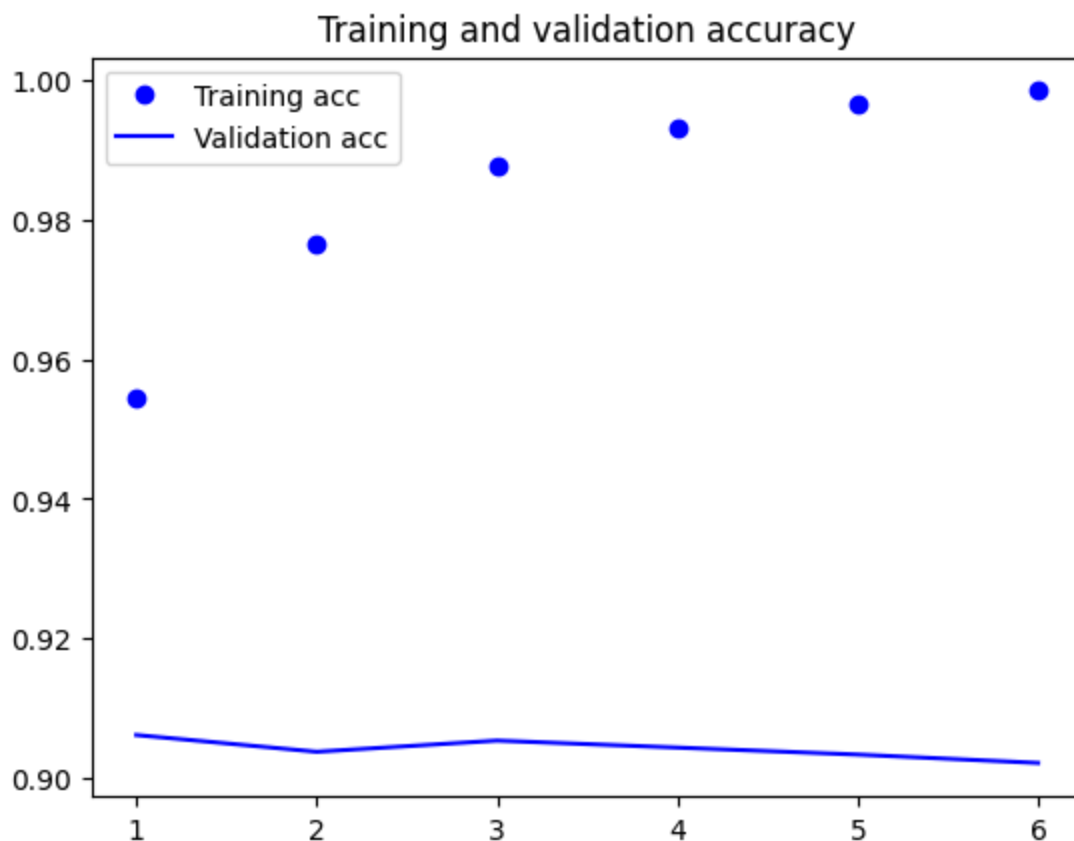
Epoch 5/200

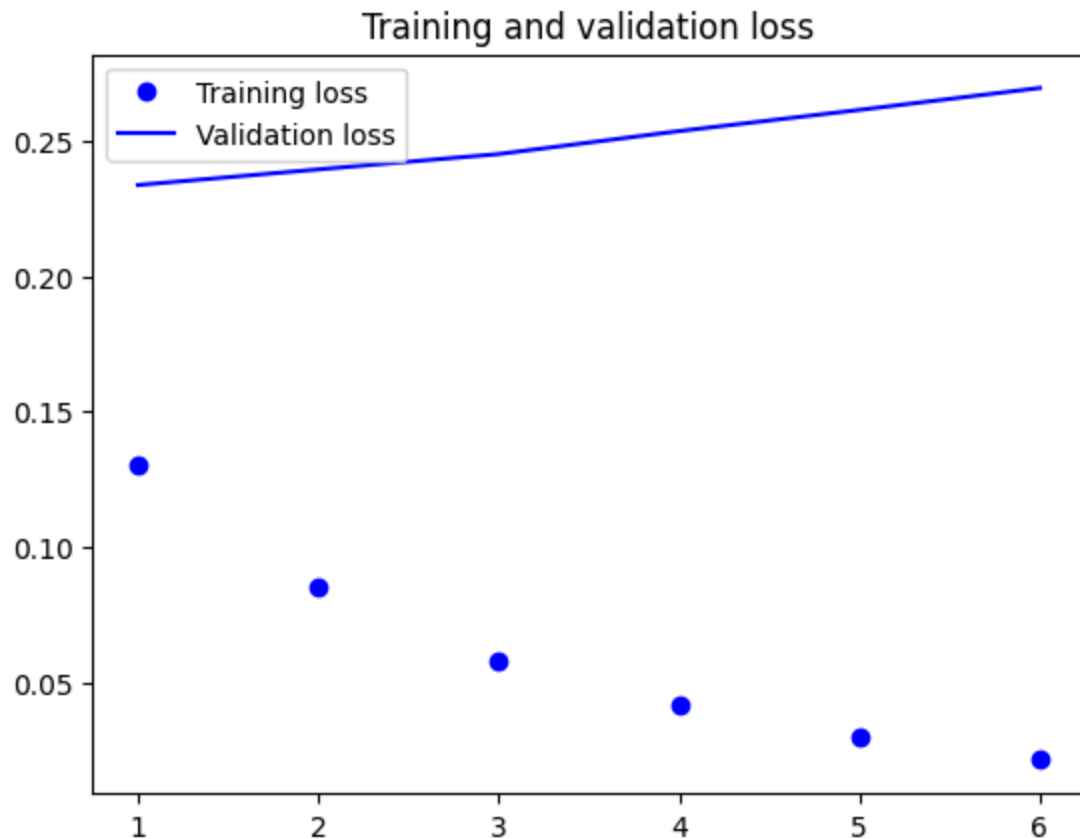
625/625 ————— **82s** 71ms/step - acc: 0.9972 - loss: 0.0278 - val_acc: 0.9034 - val_loss: 0.2614

Epoch 6/200

625/625 ————— **82s** 71ms/step - acc: 0.9991 - loss: 0.0201 - val_acc: 0.9022 - val_loss: 0.2695

```
In [ ]: plot_history(history)
```





Phase 3: Fine-tune right-branch embedding

```
In [ ]: model = keras.models.load_model('LSTM_Conv1D_retrain.keras')
        for i in range(len(model.layers)):
            model.layers[i].trainable = False
        model.layers[3].trainable = True
        model.compile(optimizer='adam',
                      loss='binary_crossentropy',
                      metrics=['acc'])
        model.summary()
```

Model: "functional"

Layer (type)	Output Shape	Param #	Connected to
input_layer (InputLayer)	(None, 500)	0	–
input_layer_1 (InputLayer)	(None, 500)	0	–
embedding (Embedding)	(None, 500, 300)	3,000,000	input_layer[0]
embedding_1 (Embedding)	(None, 500, 300)	3,000,000	input_layer_1[0]
bidirectional (Bidirectional)	(None, 500, 256)	439,296	embedding[0][0]
bidirectional_1 (Bidirectional)	(None, 500, 256)	439,296	embedding_1[0][0]
conv1d (Conv1D)	(None, 494, 64)	114,752	bidirectional[0][0]
conv1d_1 (Conv1D)	(None, 494, 64)	114,752	bidirectional_1[0][0]
global_max_pooling... (GlobalMaxPooling1D)	(None, 64)	0	conv1d[0][0]
global_max_pooling... (GlobalMaxPooling1D)	(None, 64)	0	conv1d_1[0][0]
concatenate (Concatenate)	(None, 128)	0	global_max_po global_max_po
flatten (Flatten)	(None, 128)	0	concatenate[0]
dense (Dense)	(None, 512)	66,048	flatten[0][0]
dropout (Dropout)	(None, 512)	0	dense[0][0]
dense_1 (Dense)	(None, 1)	513	dropout[0][0]

Total params: 7,174,657 (27.37 MB)

Trainable params: 3,000,000 (11.44 MB)

Non-trainable params: 4,174,657 (15.93 MB)

```
In [ ]: callbacks_list = [
        keras.callbacks.EarlyStopping(monitor='val_loss', patience=5,),
        keras.callbacks.ModelCheckpoint(filepath='LSTM_Conv1D_retrain_1.keras',
history = model.fit([x_train, x_train], y_train, shuffle=True,
                    epochs=200,
                    batch_size=32,
                    callbacks=callbacks_list,
                    validation_data=(x_val, y_val))
```

```

Epoch 1/200
625/625 ————— 47s 72ms/step - acc: 0.9769 - loss: 0.0821 - va
l_acc: 0.9084 - val_loss: 0.2350
Epoch 2/200
625/625 ————— 81s 71ms/step - acc: 0.9881 - loss: 0.0530 - va
l_acc: 0.9098 - val_loss: 0.2460
Epoch 3/200
625/625 ————— 82s 71ms/step - acc: 0.9938 - loss: 0.0356 - va
l_acc: 0.9086 - val_loss: 0.2583
Epoch 4/200
625/625 ————— 44s 71ms/step - acc: 0.9968 - loss: 0.0242 - va
l_acc: 0.9050 - val_loss: 0.2709
Epoch 5/200
625/625 ————— 44s 71ms/step - acc: 0.9980 - loss: 0.0178 - va
l_acc: 0.9064 - val_loss: 0.2815
Epoch 6/200
625/625 ————— 44s 71ms/step - acc: 0.9990 - loss: 0.0124 - va
l_acc: 0.9034 - val_loss: 0.2967

```

15. Final Test Evaluation

Load the best checkpoint and evaluate on held-out IMDB test reviews.

```
In [ ]: model = keras.models.load_model('LSTM_Conv1D_retrain_1.keras')
```

```
In [ ]: test_dir = os.path.join(imdb_dir, 'test')
labels = []
texts = []
for label_type in ['neg', 'pos']:
    dir_name = os.path.join(test_dir, label_type)
    for fname in sorted(os.listdir(dir_name)):
        if fname[-4:] == '.txt':
            f = open(os.path.join(dir_name, fname))
            texts.append(f.read())
            f.close()
            if label_type == 'neg':
                labels.append(0)
            else:
                labels.append(1)
sequences = tokenizer.texts_to_sequences(texts)
x_test = pad_sequences(sequences, maxlen=maxlen)
y_test = np.asarray(labels)
```

```
In [ ]: model.evaluate([x_test, x_test], y_test)
```

```
782/782 ————— 29s 35ms/step - acc: 0.9149 - loss: 0.2195
```

```
Out[ ]: [0.2298915982246399, 0.9093999862670898]
```

Result: test accuracy exceeds 90% using staged embedding fine-tuning with early stopping.